# Foundations of Quantum Computing

–Week 2– Explore – Study

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This week we will study mathematical prerequisites to quantum computation, building up to our first encounter with quantum information. In this week's Study section you will cover the following topics:



## 2.1 Algebra and geometry of the 2D plane

We will now review some concepts of geometry and transformations of the 2D plane. We will first review certain concepts to do with transformations of 2D real-valued vectors, as in vectors in $\mathbb{R}^2$. In doing so, we will show how these concepts would be described using the 'ket' notation that we introduced last week to describe probability vectors.

### 2.1.1 Cartesian coordinates, inner products, and norms (in 'ket' notation)

We may describe a point in the 2D plane by a vector in $\mathbb{R}^2$

$$a\,|0\rangle + b\,|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}, \qquad \text{where } a, b \in \mathbb{R}. \tag{2.1}$$

Here we use the 'ket' vector notation (or 'Dirac' notation) that we introduced last week. Here, $|0\rangle$ (pronounced 'ket 0') represents the unit vector in the direction of the positive $x$ axis, to the right of the 'origin'; and $|1\rangle$ (pronounced 'ket 1') is the unit vector in the direction of the positive $y$ axis, directly up from the 'origin':

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad\qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{2.2}$$

These vectors are known as the **standard basis vectors** for $\mathbb{R}^2$. You may have seen these same vectors written as $\boldsymbol{\hat\imath}$ and $\boldsymbol{\hat\jmath}$, or $\mathbf{e}_1$ and $\mathbf{e}_2$ : we just use a slightly different notation – and label them with '0' and '1' so that later we can make the connection to bits. So: $a\,|0\rangle + b\,|1\rangle$ are the same as standard Cartesian coordinates, just using a slightly different notation. The origin itself, we write by the vector

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \in \mathbb{R}^2 \,, \tag{2.3}$$

which we also call the **zero vector**. Note that this is different from the vector $|0\rangle$. (We will not refer very often to the zero vector, which will hopefully avoid some confusion.) These concepts are all illustrated in Figure 2.1.

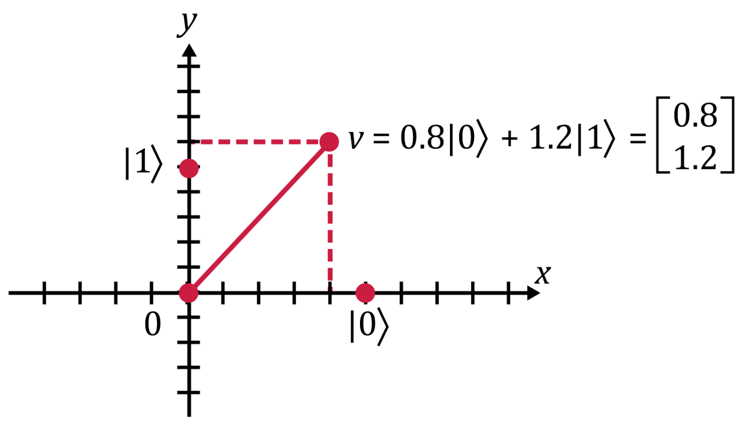


Figure 2.1: We show the location of a fairly generic point $\boldsymbol{v}$ in $\mathbb{R}^2$, represented both in 'ket' notation and as a column vector. We indicate the scale of both the $x$ and $y$ axes in increments of 0.2, with the projections of $\boldsymbol{p}$ onto the $x$ and $y$ axes to demonstrate its coordinates.
Long description: The vertical $y$ axis and horizontal $x$ axis are marked in black. The axes have short black marks regularly spaced along their lengths. Four points are marked in dark red. The origin is where the axes cross and is marked as $\mathbf{0}$; the $|0\rangle$ vector is on the positive horizontal axis; the $|1\rangle$ vector is on the positive vertical axis. The fourth point is above and to the right of the origin, and is marked as $\boldsymbol{v} = 0.8\,|0\rangle + 1.2\,|1\rangle$. Dashed light red lines are drawn perpendicularly from $\boldsymbol{v}$ to each of the axes, and a solid line connects it to the origin.

This 'ket' notation is not just a fancy way of writing the standard basis vectors — in quantum mechanics, one often writes **every** vector using 'ket' notation. In this module, however, we use a very strict convention:

**Notation:** Any single vector written as a 'ket' (for example, $|0\rangle$, $|x\rangle$, $|\psi\rangle$, and so forth), is a **unit vector**.

A 'unit vector' $\boldsymbol{v}$ is just a vector which has length 1. In $\mathbb{R}^2$, this is just the distance of the point $\boldsymbol{v}$ from the origin. We can express this more precisely in algebra (and generalise it to more than just vectors $\mathbb{R}^2$) as follows. For any two **real-valued** vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ – of any dimension, but the same dimension as each other – we define the **inner product** of $\boldsymbol{u}$ and $\boldsymbol{v}$ to be

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle \;=\; \boldsymbol{u}^\top \boldsymbol{v} \;=\; u_1 v_1 \,+\, u_2 v_2 \,+\, \cdots \,+\, u_n v_n \,, \tag{2.4}$$

where $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$. (We will generalise this definition slightly when we consider complex numbers.) We then define the **norm** (or the 'length') of $\boldsymbol{v}$ to be

$$\big\|\boldsymbol{v}\big\| \;=\; \sqrt{\langle \boldsymbol{v}, \boldsymbol{v}\rangle} \;=\; \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}\;. \tag{2.5}$$

In the case that $\boldsymbol{v} = a\,|0\rangle + b\,|1\rangle \in \mathbb{R}^2$, this means that

$$\big\|\boldsymbol{v}\big\| \;=\; \sqrt{\langle \boldsymbol{v}, \boldsymbol{v}\rangle} \;=\; \sqrt{a^2 + b^2}. \tag{2.6}$$

This might look familiar, as the description of the length of the hypothenuse, of a right-angle triangle with side-lengths $a$ and $b$ adjacent to the right angle. This describes the length of the vector $\boldsymbol{v}$, described as a location away from the origin (as illustrated in Figure 2.2.2), as $|0\rangle$ and $|1\rangle$ are at right angles to each other. This result generalises to three dimensions, and in fact, we use the value of $\|\boldsymbol{v}\|$ to **define** the notion of length for real-valued vectors of all higher dimensions. (Again, we will extend this slightly when we consider complex numbers.)

**Example** (Lengths of vectors)**:** Let's look at the lengths of some familiar vectors in $\mathbb{R}^2$.

- The zero vector $\mathbf{0}$ has norm 0: $\big\|\mathbf{0}\big\|^2 = \sqrt{0^2 + 0^2} = 0$.
- The standard basis vectors $|0\rangle$ and $|1\rangle$ have norm 1:

$$\big\|\;|0\rangle\;\big\| \;=\; \sqrt{1^2 + 0^2} \;=\; 1 \qquad \text{and} \qquad \big\|\;|1\rangle\;\big\| \;=\; \sqrt{0^2 + 1^2} \;=\; 1. \tag{2.7}$$

- The norms of $|0\rangle + |1\rangle$ and $|0\rangle - |1\rangle$ are

$$\big\|\;|0\rangle + |1\rangle\;\big\| \;=\; \sqrt{1^2 + 1^2} \;=\; \sqrt{2}\,,\quad \big\|\;|0\rangle - |1\rangle\;\big\| \;=\; \sqrt{1^2 + (-1)^2} \;=\; \sqrt{2}. \tag{2.8}$$

You can think of $\|\boldsymbol{v}\|$ as being similar to the absolute value $|z|$ of a number $z$. In fact, we describe a connection between these two concepts in Section 2.2.1.

Vectors of length 1 (or 'unit length', or 'unit norm') are so important and useful that we single them out for special attention. In addition to the term 'unit vector', we tend to think of them as a notion of **pure direction**. We can express this in terms of how to compute the 'unit vector' in the direction of a vector $\boldsymbol{v}$:

**Proposition 1** (Rescaling vectors)**.** *If $\big\|\boldsymbol{v}\big\| \neq 0$ then $\dfrac{\boldsymbol{v}}{\|\,\boldsymbol{v}\,\|} = \frac{1}{\|\boldsymbol{v}\|}\,\boldsymbol{v}$ is a unit vector.*

**Proof** The proof of this remains as an exercise. Hint: The norm of a vector is a real number, so can be treated like any other number in algebra.

**Example** (Rescaling non-unit vectors)**:** In the preceding 'lengths of vectors' example, we saw that $\big\|\;|0\rangle + |1\rangle\;\big\| = \big\|\;|0\rangle - |1\rangle\;\big\| = \sqrt{2}$. We can rescale these to make unit vectors by multiplying each of them by $\frac{1}{\sqrt{2}}$:

$$\tfrac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big)\;, \qquad\qquad \tfrac{1}{\sqrt{2}}\big(|0\rangle - |1\rangle\big)\;. \tag{2.9}$$

For a named non-zero vector $\boldsymbol{v} \in \mathbb{R}^n$, we will write it as $|\boldsymbol{v}\rangle$ for the unit vector in the direction of $\boldsymbol{v} \neq \mathbf{0}$. (This is sometimes written $\hat{\boldsymbol{v}}$.) Then, this allows us to separate any non-zero vector into a notion of length and direction:

$$\boldsymbol{v} \;=\; \big\|\boldsymbol{v}\big\| \cdot |\boldsymbol{v}\rangle \,. \tag{2.10}$$

In fact, in this module, the majority of vectors that we consider will be unit vectors. (The process of rescaling a vector to obtain a unit vector, will be more useful to us than separating a vector into its length and direction.)

In the case of real-valued unit vectors $|v\rangle \in \mathbb{R}^n$, we define a special symbol for the column vector $|v\rangle^\top$: we write

$$\langle v| \;=\; |v\rangle^\top \;. \tag{2.11}$$

(Again, we extend this slightly when we discuss complex numbers.) We say 'bra-$v$' to refer to the row vector $\langle v|$. We define this notation so that the result of multiplying a 'bra' vector with a 'ket' vector (performing a normal row-column vector multiplication) is a **bracket**, which is to say, the inner product of $|u\rangle$ and $|v\rangle$:

$$\langle u| \cdot |v\rangle \;=\; \langle u\,|v\rangle \;=\; u_1 v_1 + u_2 v_2 + \cdots + u_n v_n \;. \tag{2.12}$$

This little piece of word-play is a (very small) part of the reason for using 'ket' (or 'bra-ket', or 'Dirac') notation: the main reason is the fact that it allows us to more easily describe transformations of complicated vectors, as we saw last week when discussing randomised computation. The same notational advantage will prove helpful for quantum computation as well.

### 2.1.2 Angles, elementary trigonometry and polar coordinates

It will occasionally be useful to consider the unit vectors $|u\rangle \in \mathbb{R}^2$ in particular. These are the points on the plane which are at a distance of 1 from the origin – which is to say, a circle of radius 1, centred on the origin. This is illustrated in Figure 2.2, and is called the **unit circle**. The components of a unit vector are determined by the angle $\theta$, which is made by $|u\rangle$ with the $x$ axis, through the **trigonometric functions** $\cos(\theta)$ and $\sin(\theta)$. We will now review these notions.

**Measuring angles in radians.**
You may be accustomed to describing angles in 'degrees', with 360° in a circle, 180° representing a half-rotation or reversal in direction, 90° being a right-angle, and so forth. But in most of physics and mathematics, it is more helpful to measure angles in **radians**. This is a different unit of measurement, which is more directly helpful for computing relationships between angles and lengths. We can convert between these using the formula

$$\langle\text{angle in radians}\rangle = \frac{\pi}{180} \times \langle\text{angle in degrees}\rangle \tag{2.13}$$

where $\pi = 3.1415926\ldots$. So, a full revolution is an angle of $2\pi$, a half-rotation is an angle of $\pi$, and a right-angle is an angle of $\frac{\pi}{2}$. (See Table 1 for a quick conversion table.)

| Degrees | 0° | 22.5° | 30° | 45° | 60° | 90° | 120° | 135° | 150° | 180° | 270° | 360° |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Radians | 0 | $\pi/8$ | $\pi/6$ | $\pi/4$ | $\pi/3$ | $\pi/2$ | $2\pi/3$ | $3\pi/4$ | $5\pi/6$ | $\pi$ | $3\pi/2$ | $2\pi$ |

Table 1: A quick degrees-to-radians conversion table.

**Angles with direction.**
In many cases, we will think of an angle as being in a particular **direction**: clockwise or anticlockwise. The usual convention is for a **positive** angle to represent a anticlockwise rotation, and for a **negative** angle to represent a clockwise rotation. Figure 2.2 illustrates the way that we use angles to indicate the direction of a point from the origin in the 2D plane.

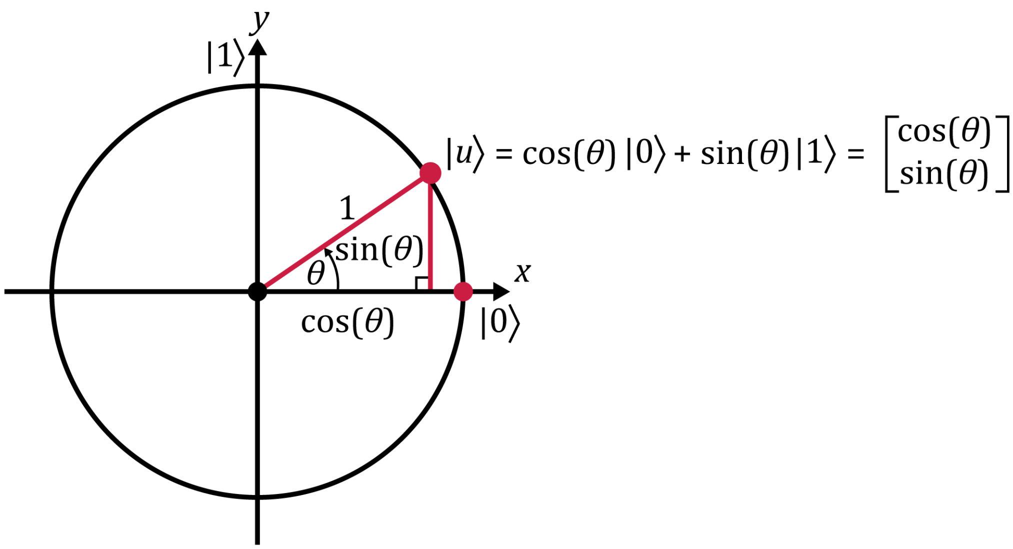

Figure 2.2: The components of a unit vector $|u\rangle \in \mathbb{R}^2$ are given by the trigonometric functions $\cos(\theta)$ and $\sin(\theta)$, where $\theta$ is the anticlockwise angle made between the $x$ axis and the vector $|u\rangle$. Also illustrated is the right-angle triangle formed by the two components of $|u\rangle$. Tracing out the locations of all such unit vectors (for various values of $\theta$ from 0 to $2\pi$) describes a circle of radius 1 about the origin, called the 'unit circle'.
Long description: The horizontal $x$ axis and vertical $y$ axis are marked in black, with points marked on them indicating the vectors $|0\rangle$ and $|1\rangle$. A point is marked in dark red in the upper right at the same distance from the origin as $|0\rangle$ and $|1\rangle$. Next to that point is the label $|u\rangle = \cos(\theta)\,|0\rangle + \sin(\theta)\,|1\rangle$, and a column vector with the coefficients $\cos(\theta)$ and $\sin(\theta)$. The line from the origin to that point is also marked in dark red, with the number 1 written above it. The vertical line below the marked point meets a horizontal line from the origin to form the right-angle of a triangle (marked in light red). The angle at the origin of this triangle is labelled $\theta$, the marking to indicate this angle includes a small arrow to reinforce the convention that this angle is measured anticlockwise from the positive $x$ axis. The horizontal and vertical lines of the triangle are marked $\cos(\theta)$ and $\sin(\theta)$ respectively. The circle of all points, which are the same distance from the origin as $|0\rangle$, $|1\rangle$, and $|u\rangle$, is indicated in dark red.

**Trigonometry.**
In this module, we will not use a lot of complicated trigonometry – but basic trigonometry is absolutely essential. In particular, we rely heavily on the 'cosine' function $\cos(\theta)$ and the 'sine' function $\sin(\theta)$ for angles $\theta \in \mathbb{R}$. (We usually use the Greek letter '$\theta$' to represent angles, particularly in the 2D plane: often this angle is measured as an angle anticlockwise from the positive $x$ axis.) As $\theta$ moves from 0 (pointing right) to $\pi/2$ (pointing up) to $\pi$ (pointing left) to $3\pi/2$ (pointing down), it eventually comes back round to $2\pi$ (pointing right again). We generally ignore full revolutions of $2\pi$ when describing angles; so you'll usually see angles expressed as either $0 \leq \theta < 2\pi$ or sometimes as $-\pi < \theta \leq \pi$.
Figure 2.3 is a plot of $\sin(\theta)$ and $\cos(\theta)$ as $\theta$ ranges from 0 to $2\pi$. The curves a e identical, apart from a horizontal shift of $\pi/2$ relating one to the other, as $\cos(\theta) = \sin\left(\theta + \frac{\pi}{2}\right)$.

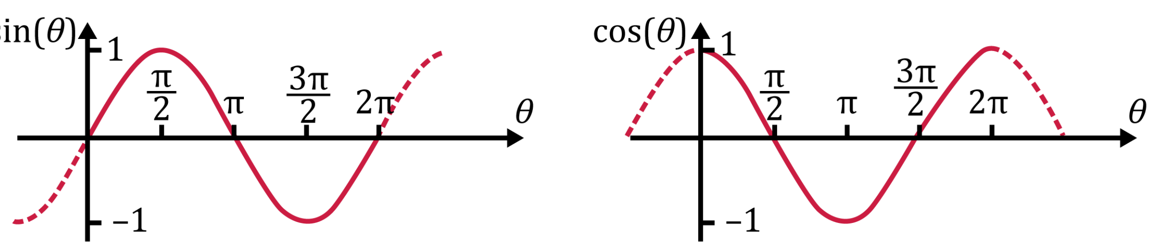

Figure 2.3: These two graphs show $\theta$ in the horizontal axis, and then respectively a plot of $\sin(\theta)$ and $\cos(\theta)$. These functions are both **periodic**: they repeat after every left or right shift by $2\pi$. We can then pass in any real number as a value of $\theta$ and the resulting value of $\sin(\theta)$ or $\cos(\theta)$ can be calculated by continuing the pattern as shown by the dashed lines.
Long description: Two figures, very alike except that the plotted curve (in dark red) starts at a different point in the pattern as described below. The axes are in black, the vertical axis is marked at 1 and $-1$. The horizontal axis is marked in increments of $\pi/2$. The curve in the left hand figure (for $\sin(\theta)$), ignoring the dashed section, starts at the origin by heading up and right, smoothly cresting at a height of 1 (for $\theta = \pi/2$), crossing the horizontal axis at $\theta = \pi$, curving down to reach a lowest value of $-1$ at $\theta = 3\pi/2$, and then heading back up to the horizontal axis to reach at at $\theta = 2\pi$. The corresponding curve for $\cos(\theta)$ starts at a height of 1 when $\theta = 0$, curving down through the horizontal axis to reach a low of $-1$ when $\theta = \pi$, then curves back up again to reach a height of 1 when $\theta = 2\pi$. For both graphs there are dashed lines which trace out a continuation of these curves beyond the ranges of $\theta = 0$ and $\theta = 2\pi$.

**The 'circle identity'.**
We will not require you to memorise many trigonometric formulas, and will instead inform/remind you of any special 'trignometric identities' that you might need to solve problems. One formula that we **do** expect you to remember, is the following equation which connects $\cos(\theta)$ and $\sin(\theta)$ to coordinates in the unit circle:

$$\sin^2(\theta) + \cos^2(\theta) = 1 \,. \tag{2.14}$$

(Here, '$\sin^2(\theta)$' means the same as $[\sin(\theta)]^2$, and '$\cos^2(\theta)$' means the same as $[\cos(\theta)]^2$. The notation shown in Eqn. (2.14) might seem unusual to some readers, but it is very common, and we will use it throughout the module.)

**Polar coordinates.**
In the 2D plane, the angle $\theta$ is in this case all the information we need to specify a direction. Specifically, every unit vector $|u\rangle \in \mathbb{R}^2$ can be written as

$$|u\rangle \;=\; \cos(\theta)\,|0\rangle + \sin(\theta)\,|1\rangle \tag{2.15}$$

for some angle $0 \leqslant \theta < 2\pi$ (or $-\pi < \theta \leqslant \pi$): this is also illustrated in Figure 2.2. This means that we can represent any non-zero vector $\boldsymbol{v} \in \mathbb{R}^2$, as a distance $r = \lVert \boldsymbol{v} \rVert$ from the origin, and a direction $\theta$ corresponding to the unit vector $|\boldsymbol{v}\rangle = \frac{1}{r}\boldsymbol{v}$. We describe the values of $(r, \theta)$ of this kind, as the **polar coordinates** of $\boldsymbol{v}$. From the description above, for $r = \lVert \boldsymbol{v} \rVert$ and $|\boldsymbol{v}\rangle = \frac{1}{r}\boldsymbol{v} = \cos(\theta)\,|0\rangle + \sin(\theta)\,|1\rangle$, we can see how to convert from polar coordinates to Cartesian coordinates:

$$\begin{array}{ccccl} \text{Polar} & \to & \text{Cartesian} & & \\ (r,\theta) & \mapsto & \begin{bmatrix} r\cos(\theta) \\ r\sin(\theta) \end{bmatrix} & = & r\cos(\theta)\,|0\rangle \;+\; r\sin(\theta)\,|1\rangle\,. \end{array} \tag{2.16}$$

(We will generally not require you to make the conversion from Cartesian coordinates to polar coordinates, except in easy special cases: see Figure 2.) Polar coordinates can be very helpful on occasion to describe operations on the plane, particularly when they preserve the unit circle – in which case, the operation affects just the direction angle $\theta$.

| Ket notation | $\lvert 0\rangle$ | $\frac{1}{\sqrt{2}}\big(\lvert 0\rangle + \lvert 1\rangle\big)$ | $\lvert 1\rangle$ | $\frac{1}{\sqrt{2}}\big(-\lvert 0\rangle + \lvert 1\rangle\big)$ | $-\lvert 0\rangle$ | $-\lvert 1\rangle$ | $\frac{1}{\sqrt{2}}\big(\lvert 0\rangle - \lvert 1\rangle\big)$ |
|---|---|---|---|---|---|---|---|
| Polar angle $\theta$ | $0$ | $\pi/4$ | $\pi/2$ | $3\pi/4$ | $\pi$ | $-\pi/2$ | $-\pi/4$ |

Table 2: Values of the polar angle $\theta$ for some unit vectors in the 2D plane. The angle values shown range as $-\pi < \theta \leqslant \pi$: where these values are negative, equivalent values $0 \leqslant \theta < 2\pi$ can be obtained simply by adding $2\pi$ to the value of $\theta$ given in the table.

### 2.1.3 Linear transformations of the 2D plane

The reason for using vectors to describe points in the 2D plane, is because we are interested in **linear transformations** of the plane. For operations on the plane, these are functions $f : \mathbb{R}^2 \to \mathbb{R}$ or $f : \mathbb{R}^2 \to \mathbb{R}^2$ which are nicely behaved, in the following sense:

- For any 'scalar' $a \in \mathbb{R}$ and any $\boldsymbol{v} \in \mathbb{R}^2$, we have $f(a\boldsymbol{v}) = af(\boldsymbol{v})$. (We say that we can 'pull in' or 'pull out' scalar factors through $f$.)
- For any two vectors $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^2$, we have $f(\boldsymbol{v} + \boldsymbol{w}) = f(\boldsymbol{v}) + f(\boldsymbol{w})$. (We say that $f$ 'distributes over sums'.)

Together, this implies that $f$ distributes over **linear combinations**: weighted sums of vectors. For any scalars $a_1, a_2, \ldots, a_n \in \mathbb{R}$ and vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_n \in \mathbb{R}^m$, we have

$$f\big(a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n\big) \;=\; a_1 f(\boldsymbol{v}_1) + a_2 f(\boldsymbol{v}_2) + \cdots + a_n f(\boldsymbol{v}_n)\;. \tag{2.17}$$

More generally, we can consider linear transformations of vectors $\boldsymbol{v} \in \mathbb{R}^m$ of any dimension: these are just functions $f : \mathbb{R}^m \to \mathbb{R}^n$ which have the same properties described above. We've already seen examples of this as well, in the effect of `RAALcr` instructions on probability distributions.

For a linear transformation (which we may sometimes describe as a 'linear operation' or 'linear map'), we can determine what it does to all vectors, just from what it does to the standard basis. Suppose that $f\big(\lvert 0\rangle\big) = \boldsymbol{u}_0$ and $f\big(\lvert 1\rangle\big) = \boldsymbol{u}_1$ for some pair of vectors $\boldsymbol{u}_0, \boldsymbol{u}_1 \in \mathbb{R}^2$. Then for a vector $\boldsymbol{v} = a_0\lvert 0\rangle + a_1\lvert 1\rangle$, we have

$$f(\boldsymbol{v}) \;=\; f\Big(a_0\lvert 0\rangle + a_1\lvert 1\rangle\Big) \;=\; a_0\, f\big(\lvert 0\rangle\big) \;+\; a_1\, f\big(\lvert 1\rangle\big) \;=\; a_0\,\boldsymbol{u}_0 \;+\; a_1\,\boldsymbol{u}_1\;. \tag{2.18}$$

This is an important property about linear functions: if we describe how they act on the standard basis vectors, the way they behave on all other vectors follows 'by linearity' – extending that description to all linear combinations of the standard basis.

Not **all** of the operations we do to vectors are linear transformations. We have already seen examples of two functions which are **not** linear: the norm map $\boldsymbol{v} \mapsto \big\|\boldsymbol{v}\big\|$ does not satisfy these properties, nor does the rescaling map $\boldsymbol{v} \mapsto \frac{1}{\|\boldsymbol{v}\|}\boldsymbol{v}$. But many of the operations which we will see in this course, **are** linear transformations.

An important fact about linear functions, is how we would tend to compute with them in practice, which is by using matrices:

**Proposition:** A function $f : \mathbb{R}^m \to \mathbb{R}^n$ is linear, if and only if $f(\boldsymbol{v}) = M\boldsymbol{v}$ for some $m \times n$ matrix $M$.

The reason why we introduce a distinction between a 'linear transformation', and a matrix, is that it is sometimes important to distinguish between a pure 'function' on the one hand (such as a rotation of the plane, or the transformation of probability vectors realised by some `RAALcr` procedure), and a specific way of **representing** that function on the other (by a matrix). It will soon become important to make distinctions between matrices/vectors, and the things which those matrices **represent** – because even when that representation is very useful, it is not always one-to-one.

Once again, we will revisit some of these notions of linearity when we introduce complex numbers. For now, we proceed by considering specific important examples on the 2D plane. The most important operations on $\mathbb{R}^2$ for us, are **rotations**, **reflections** and **projections**. (We will see how to generalise these operations later in the module, but in fact these operations on the 2D plane specifically will be important later.)

A necessary result of a linear map $f : \mathbb{R}^2 \to \mathbb{R}^2$, is that $f(\mathbf{0}) = \mathbf{0}$). This is because, for any vector $\boldsymbol{v} \in \mathbb{R}^2$ (and where we let $\boldsymbol{w} = f(\boldsymbol{v}) \in \mathbb{R}^2$), we have

$$f(\mathbf{0}) \;=\; f\big(0 \cdot \boldsymbol{v}\big) \;=\; 0 \cdot f(\boldsymbol{v}) \;=\; 0 \cdot \boldsymbol{w} \;=\; \mathbf{0}. \tag{2.19}$$

What this means is that all of the examples that we consider below – of rotations, reflections, and projections – all will preserve the origin. Every rotation operation will be a rotation of the 2D plane around the origin, and every reflection or projection operation will be with respect to a line or a set of axes which pass through the origin (something that from now on we will tend to assume about a set of axes).

The fact that such rotations, projections, and reflections are linear transformations, is something that you should remember and also apply when useful to do so.

### (a) Projections in $\mathbb{R}^2$

A **projection** (or a **projector**) is a linear transformation which we define with respect to a pair of axes, possibly different from the $x$ and $y$ axes. We are interested in **orthogonal projections**, involving axes which are at right angles (in other words, 'orthogonal') to each other in the plane.

An orthogonal projection takes a vector $\boldsymbol{v}$, and maps it to the 'shadow' that it casts on one of the axes – removing any part of $\boldsymbol{v}$ which is parallel to the other axis (this is illustrated in Figure 2.4. Next we will describe these ideas more precisely.

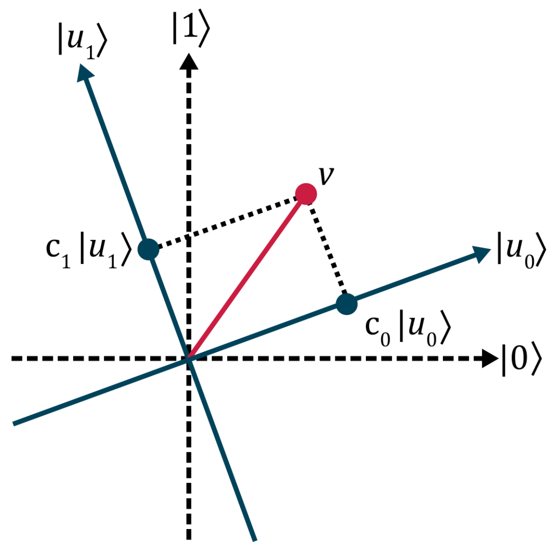


Figure 2.4: Projections of a vector in the plane onto orthogonal axes. For a generic vector $\boldsymbol{v} \in \mathbb{R}^2$, we can not only describe it as being constructed as a vector $\boldsymbol{v} = v_0|0\rangle + v_1|1\rangle$ in terms of its coefficients as a column vector, but we can also consider its components $c_0|u_0\rangle$ and $c_1|u_1\rangle$ with respect to any other basis of orthogonal unit vectors, $|u_0\rangle$ and $|u_1\rangle$. These components $c_0\,|u_0\rangle$ and $c_1\,|u_1\rangle$ are then the projections of $\boldsymbol{v}$, onto the axes described by the unit vectors $|u_0\rangle$ and $|u_1\rangle$ respectively.
Long description: A diagram of the 2D plane. The horizontal and vertical axes are labeled by $|0\rangle$ and $|1\rangle$, and are drawn as dashed lines. Another set of orthogonal axes are drawn as solid lines, rotated anticlockwise from the horizontal and vertical axes by a modest angle. This second set of axes is labeled by unit vectors $|u_0\rangle$ and $|u_1\rangle$. A vector $\boldsymbol{v}$ in the plane is shown, with a line showing its displacement from the origin. Projections from $\boldsymbol{v}$ onto the $|u_0\rangle$ and $|u_1\rangle$ axes are shown, and labeled as vectors $c_0|u_0\rangle$ and $c_1|u_1\rangle$ respectively.

We can describe an axis in the 2D plane by a unit vector $|u\rangle$, which points in the same direction as the axis. When we want to consider two **orthogonal** axes, we then consider two unit vectors $|u_0\rangle$ and $|u_1\rangle$ which are orthogonal, so that they point at right angles to one another. This means that $\langle u_0\,|u_1\rangle = 0$, which we will be able to use to do actual calculations when we consider the projections onto the '$|u_0\rangle$ axis' and the '$|u_1\rangle$ axis'.

When we have two such unit vectors, we can describe projection operations much as we did in Week 1 for the vectors $|0\rangle$ and $|1\rangle$:

$$|u_0\rangle\,\langle u_0|\;, \qquad\qquad |u_1\rangle\,\langle u_1|\;. \tag{2.20}$$

We can then describe any vector $\boldsymbol{v}$, as a linear combination of $|u_0\rangle$ and $|u_1\rangle$, using the coefficients

$$c_0 \;=\; \langle u_0|\;\boldsymbol{v}\;, \qquad\qquad c_1 \;=\; \langle u_1|\;\boldsymbol{v}\;, \tag{2.21}$$

writing $\boldsymbol{v} = c_0 |u_0\rangle + c_1 |u_1\rangle$. Then the operator $|u_0\rangle \langle u_0|$ can be thought of as 'selecting' for the component $c_0 |u_0\rangle$ of $\boldsymbol{v}$, that aligns with the $|u_0\rangle$ axis:

$$\begin{aligned} |u_0\rangle \langle u_0| \cdot \boldsymbol{v} \;&=\; |u_0\rangle \langle u_0| \Big( \underline{c_0 |u_0\rangle} + c_1 |u_1\rangle \Big) \\ &=\; c_0 |u_0\rangle \langle u_0 | u_0\rangle + c_1 |u_0\rangle \langle u_0 | u_1\rangle \;=\; \underline{c_0 |u_0\rangle} \,. \end{aligned} \tag{2.22a}$$

Similarly, the operator $|u_1\rangle \langle u_1|$ can be thought of as 'selecting' for the component $c_1 |u_1\rangle$ of $\boldsymbol{v}$, that aligns with the $|u_1\rangle$ axis:

$$\begin{aligned} |u_1\rangle \langle u_1| \cdot \boldsymbol{v} \;&=\; |u_1\rangle \langle u_1| \Big( c_0 |u_0\rangle + \underline{c_1 |u_1\rangle} \Big) \\ &=\; c_0 |u_1\rangle \langle u_1 | u_0\rangle + c_1 |u_1\rangle \langle u_1 | u_1\rangle \;=\; \underline{c_1 |u_1\rangle} \,. \end{aligned} \tag{2.22b}$$

Note that performing a projection twice has no further effect beyond performing it once. If you project $\boldsymbol{v}$ onto the $|u_0\rangle$ axis, and remove the component of $\boldsymbol{v}$ along the $|u_1\rangle$ axis, there is no more $|u_1\rangle$-component to remove if you project it a second time. Then, for a projection operation $P$, we have $P^2 = P$. For an orthogonal projection in the 2D plane onto a '$|u\rangle$ axis', we can represent this operation just using the $2 \times 2$ matrix $|u\rangle \langle u|$, without even specifying the other orthogonal axis involved. We can then describe these projections in $\mathbb{R}^2$ as operators $P$, such that $P^2 = P$ and $P^\top = P$. (Again, when we come to consider complex numbers, we will extend this notion slightly.)

### (b) Reflections in $\mathbb{R}^2$

A **reflection** is a linear transformation which is similar in some respects to a projection. These are also defined with respect to a pair of (orthogonal) axes, represented by unit vectors $|u_0\rangle$ and $|u_1\rangle$. A 'reflection through the $|u_0\rangle$ axis' is then a transformation which takes a vector $\boldsymbol{v}$, and 'flips' the component of $\boldsymbol{v}$ which is aligned with the $|u_1\rangle$ axis – leaving the $|u_0\rangle$ component of $\boldsymbol{v}$ unchanged. This is illustrated in Figure 2.5.

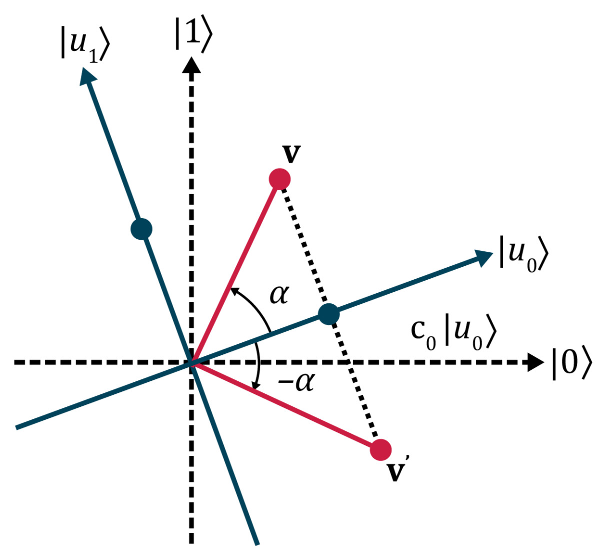

Figure 2.5: Reflection of a 2D vector through an axis. For a pair of orthogonal unit vectors $|u_0\rangle$ and $|u_1\rangle$ describing a pair of axes, and a vector $\boldsymbol{v} = c_0\,|u_0\rangle + c_1\,|u_1\rangle$, we may describe the reflection of $\boldsymbol{v}$ through the $|u_0\rangle$ axis as leaving the component $c_0\,|u_0\rangle$ unchanged, and changing the component $c_1\,|u_1\rangle$ to point in the opposite direction, yielding a vector $\boldsymbol{v}' = c_0\,|u_0\rangle - c_1\,|u_1\rangle$. If the angle made by the vector $\boldsymbol{v}$ with respect to the $|u_0\rangle$ axis is $\alpha$, the reflection $\boldsymbol{v}'$ then forms an angle of $-\alpha$ with the same axis.
Long description: A diagram of the 2D plane. The horizontal and vertical axes are labeled by $|0\rangle$ and $|1\rangle$, and are drawn as dashed lines. Another set of orthogonal axes are drawn as solid lines, rotated anticlockwise from the horizontal and vertical axes by a modest angle. This second set of axes is labeled by unit vectors $|u_0\rangle$ and $|u_1\rangle$. A vector $\boldsymbol{v}$ in the plane is shown, with a line showing its displacement from the origin. The projection from $\boldsymbol{v}$ onto the $|u_0\rangle$ axis is shown, and labeled by vectors $c_0|u_0\rangle$. The angle between the $|u_0\rangle$ axis and the vector $\boldsymbol{v}$ is indicated as $\alpha$. A second vector, the reflection of $\boldsymbol{v}$ through the $|u_0\rangle$ axis, is labeled as $\boldsymbol{v}'$. The displacement of $\boldsymbol{v}'$ from the origin is indicated by a dashed line. The angle between this vector and the $|u_0\rangle$ axis is indicated as $-\alpha$.

To be precise, what it means to 'flip' the $|u_1\rangle$ component of a vector, is to multiply that one component by $-1$:

$$c_0\,|u_0\rangle + c_1\,|u_1\rangle \quad \xmapsto{\text{reflection through } |u_0\rangle} \quad c_0\,|u_0\rangle - c_1\,|u_1\rangle\ . \tag{2.23}$$

('Selecting' a component of a vector $\boldsymbol{v}$, and changing it independently of the others, is possible precisely because this is a linear transformation.)

Reflection operations do not change the lengths of vectors: only the directions in which they point. That is, reflections can be described purely in terms of how they affect the polar angles of a vector $\boldsymbol{v}$: specifically, by changing the angle $\alpha$ that $\boldsymbol{v}$ makes with the $|u_0\rangle$ axis. Any anticlockwise angle that is formed between $\boldsymbol{v}$ and the $|u_0\rangle$ axis, is transformed into a clockwise angle, and vice-versa (as shown in Figure 2.6).

Let $\theta_0$ be the polar angle of the vector $|u_0\rangle$. For some other non-zero vector $\boldsymbol{v}$, let $\theta$ be its polar angle, and $\alpha = \theta - \theta_0$ be the angle that $\boldsymbol{v}$ makes with respect to $|u_0\rangle$. The effect of reflecting $\boldsymbol{v}$ through the $|u_0\rangle$ axis is to 'negate' the angle $\alpha$ made between $\boldsymbol{v}$ and $|u_0\rangle$. Specifically, suppose $\boldsymbol{v}'$ is the reflection of $\boldsymbol{v}$ in the $|u_0\rangle$ axis. Let $\theta'$ be the polar angle that $\boldsymbol{v}'$ makes with the $x$ axis, and let $\alpha' = \theta' - \theta_0$ be the angle that $\boldsymbol{v}'$ makes with the $|u_0\rangle$ axis. Then the effect of reflecting $\boldsymbol{v}$ in the $|u_0\rangle$ axis to get $\boldsymbol{v}'$, is described by

$$\alpha' = -\alpha\ , \qquad \text{or equivalently,} \qquad (\theta' - \theta_0) \ = \ -(\theta - \theta_0). \tag{2.24}$$

Simplifying the above, reflection through the $|u_0\rangle$ axis can be described by the transformation of polar angles, from an initial angle of $\theta$ to a final angle of $\theta' = 2\theta_0 - \theta$. We can then describe the effect of this rotation using polar coordinates, as

$$(r, \theta) \quad \xmapsto{\text{reflection through } |u_0\rangle} \quad (r, 2\theta_0 - \theta). \tag{2.25}$$

As a special case: a reflection through $|u_0\rangle = |0\rangle$, is a reflection through the $x$ axis. This can be described by negating the polar angle, $(r, \theta) \mapsto (r, -\theta)$, illustrated in Figure 2.6.

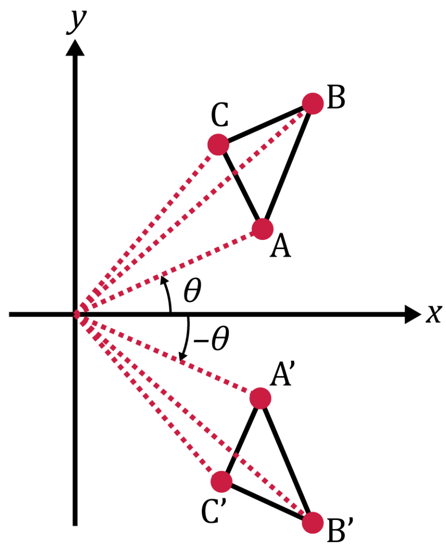


Figure 2.6: Reflection of a triangle in the plane through the $x$ axis. An illustration of a triangle $\triangle$ABC in the plane, all of whose points are to the right and upwards from the origin. The angle $\theta$, formed by the $x$ axis, the origin and the point A, is shown. A second triangle $\triangle$A$'$B$'$C$'$ is also shown, which is the result of reflecting each of the points A, B, and C through the $x$ axis. The angle $-\theta$, formed by the $x$ axis, the origin and the point A$'$, is shown.
Long description: A drawing of two triangles in a 2D plane, with several points and angles labeled. One triangle, $\triangle$ABC, appears in the upper-left quadrant. The lines formed by each of the points A, B, and C from the origin are shown as dotted lines. The anticlockwise angle formed by the line from the origin to A, and the $x$ axis, is indicated as an angle $\theta$: the anticlockwise direction represented by $\theta$ is shown by an arrow. A similar triangle (more precisely: a congruent triangle) $\triangle$A$'$B$'$C$'$ appears in the lower-right quadrant. The lines formed by each of the points A$'$, B$'$, and C$'$ from the origin are also shown as dotted lines. The point A$'$ is indicated to be the image of A after a reflection in the $x$ axis: to be specific, the polar angle formed by A$'$ and the origin with respect to the $x$ axis is described as $-\theta$. The points B$'$ and C$'$ are similarly reflected images of B and C through the $x$ axis.

### (c) Rotations in $\mathbb{R}^2$

By comparison to projections and reflections, a **rotation** in the 2D plane is straightforward to describe using polar coordinates. Put simply: a rotation by an angle $\alpha \in \mathbb{R}$, is a map which preserves lengths and adds some fixed angle to the polar angles of all points in the plane. That is, the effect for a rotation by $\alpha$ on polar coordinates, as

$$(r, \theta) \xmapsto{\text{rotation by } \alpha} (r, \theta + \alpha). \tag{2.26}$$

This is illustrated in Figure 2.7.

To describe the effect of rotations on vectors in the plane, we can use the conversion from polar coordinates to Cartesian coordinates – and also, the fact that $|1\rangle$ is itself rotated by a right angle ($\frac{\pi}{2}$ radians) from $|0\rangle$. The vectors $|0\rangle$ and $|1\rangle$ are unit vectors, and so have length 1. The polar angle of $|0\rangle$ is, by definition, zero: rotating it by $\alpha$ will result in a point with polar coordinates $(1, \alpha)$, which in Cartesian coordinates is described by the vector $\cos(\alpha)\,|0\rangle + \sin(\alpha)\,|1\rangle$. The vector $|1\rangle$ has the polar coordinates $(1, \frac{\pi}{2})$: then rotating this by $\alpha$ gives a point with coordinates $(1, \frac{\pi}{2}{+}\alpha)$. Translating this into Cartesian coordinates gives $\cos\!\big(\alpha{+}\frac{\pi}{2}\big)\,|0\rangle + \sin\!\big(\alpha + \frac{\pi}{2}\big)\,|1\rangle$. We can then make use of the following simplifications:

$$\cos\!\big(\alpha{+}\tfrac{\pi}{2}\big) = -\sin(\alpha), \qquad\qquad \sin\!\big(\alpha{+}\tfrac{\pi}{2}\big) = \cos(\alpha). \tag{2.27}$$

We then have the following descriptions of the effect of this rotation on the standard basis vectors $|0\rangle$ and $|1\rangle$, which is enough to determine how the rotation would affect all other vectors in $\mathbb{R}^2$:

$$|0\rangle \quad \xmapsto{\text{rotation by } \alpha} \quad \begin{bmatrix} \cos(\alpha) \\ \sin(\alpha) \end{bmatrix}, \qquad\qquad |1\rangle \quad \xmapsto{\text{rotation by } \alpha} \quad \begin{bmatrix} -\sin(\alpha) \\ \cos(\alpha) \end{bmatrix}. \tag{2.28}$$

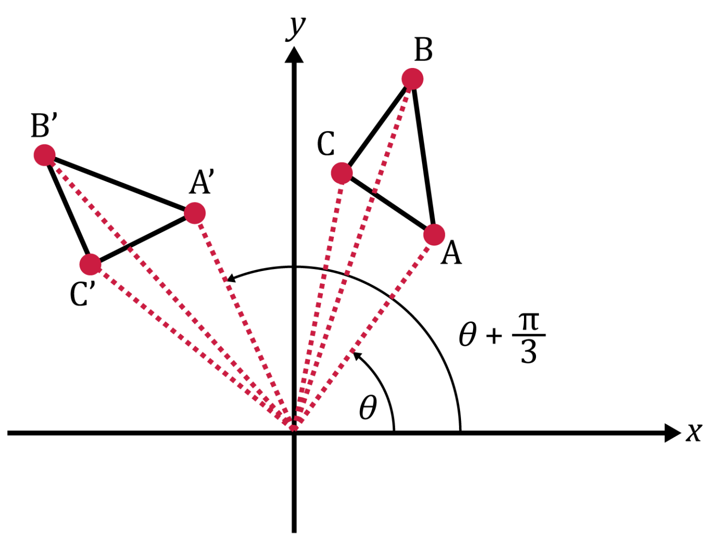

Figure 2.7: Rotation of a triangle in the plane by $\pi/3$. An illustration of a triangle $\triangle$ABC in the plane, all of whose points are to the right and upwards from the origin. The angle $\theta$, formed by the $x$ axis, the origin and the point A, is shown. A second triangle $\triangle$A′ B′ C′ is also shown, which is the result of rotating each of the points A, B, and C by an angle of $\pi/3$ (**i.e.**, by 60°) anticlockwise around the origin. The angle $\theta + \frac{\pi}{3}$, formed by the $x$ axis, the origin and the point A′, is shown.
Long description: A drawing of two triangles on a 2D plane, with several points and angles labelled. One triangle, $\triangle$ABC, appears in the upper-left quadrant. The lines formed by each of the points A, B, and C from the origin are shown as dotted lines. The anticlockwise angle formed by the line from the origin to A, and the $x$ axis, is indicated as an angle $\theta$: the anticlockwise direction represented by $\theta$ is shown by an arrow. A similar triangle (more precisely: a congruent triangle) $\triangle$A′B′C′ appears in the upper-right quadrant. The lines formed by each of the points A′, B′, and C′ from the origin are also shown as dotted lines. The point A′ is indicated to be the image of A after a anticlockwise rotation by $\frac{\pi}{2}$: to be specific, the anticlockwise angle formed, by the line from the origin to A′ and the $x$ axis, is shown to be $\theta + \frac{\pi}{2}$, which is an additional angle of $\frac{\pi}{2}$ to the angle formed by the point A relative to the $x$ axis. The points B′ and C′ are similarly rotated images of B and C respectively, by rotation about the origin by an angle of $\pi/3$.

### 2.1.4 Summary: transformations of the plane

We have seen how to represent points on the 2D plane, in Cartesian coordinates (using column vectors and ket notation) and in polar coordinates. This has helped us to describe some important transformations of the 2D plane: projections, reflections, and rotations. We will now generalise a number of these ideas further below, for higher dimensions – and for vectors whose coefficients are not all real-valued.

## 2.2 Basics of complex numbers

Complex numbers get an unfortunate reputation due to their name. They are **complex** in that they are built of more than one part. They are not any more **complicated** than vectors in $\mathbb{R}^2$, or even just polynomials, as we will now see.

A complex number is expressed as the sum of two components,

$$a + bi \tag{2.29}$$

where $a$ and $b$ are real numbers, and $i$ is the **imaginary unit**. For a complex number $a + bi$, we call $a = \mathrm{Re}(a + bi)$ the **real** part and $b = \mathrm{Im}(a + bi)$ the **imaginary** part. We write $\mathbb{C}$ to represent the set of all complex numbers.

The main thing is not to worry about what numbers such as $a + bi$ 'mean' – they don't represent a quantity of something, for example – but to consider how you can reason about them. We treat the imaginary unit $i$, like a variable such as $x$ or $y$. We can then add complex numbers together:

$$2i + 5i - 3i = 4i \qquad\qquad 0i = 0 \tag{2.30}$$

$$(2{+}i) + (-3{+}4i) = -1 + 5i \qquad\qquad (2{+}0i) + (0{+}1i) = 2 + i \tag{2.31}$$

We can also multiply complex numbers together:

$$(2i)\,(5i)\,(-3i) = -30i^3 \qquad\qquad (0)\,(i) = 0 \tag{2.32}$$

$$(2{+}i)\,(-3{+}4i) = -6 + 5i + 4i^2 \qquad\qquad (2{+}0i)\,(0{+}1i) = 2i \tag{2.33}$$

One extra piece of information that we can use, and which is the reason for defining the complex numbers in the first place, is that $i^2 = -1$ (a property no 'real' number has):

$$i^2 = -1 \qquad\qquad i^3 = i^2\, i = -i \tag{2.34}$$

$$i^4 = i^2\, i^2 = 1 \qquad\qquad i^5 = i^4\, i = i \tag{2.35}$$

$$-6 + 5i + 4i^2 = -10 + 5i \qquad\qquad -30i^3 = 30i \tag{2.36}$$

Whenever we add or multiply complex numbers together, we can use the property $i^2 = -1$ to simplify the result: and the simplified answer will then be of the form $a + bi$.

### 2.2.1 Representing complex numbers on the 2D plane

It takes the same amount of information to specify a complex number $a + bi \in \mathbb{C}$ or a point in the plane $a\,|0\rangle + b\,|1\rangle \in \mathbb{R}^2$. We can then express nearly everything about complex numbers, in terms of functions on $\mathbb{R}^2$.

To plot complex numbers in the 2D plane, we use the map

$$a + bi \;\mapsto\; \begin{bmatrix} a \\ b \end{bmatrix} \;. \tag{2.37}$$

When we talk about 'representing a complex number in the plane' we mean applying this map. (This map is invertible; there is a 1-1 correspondence between the complex numbers and the points in the plane.) This will provide us with ways of reasoning about complex numbers in terms of points in the plane (see Figure 2.8).

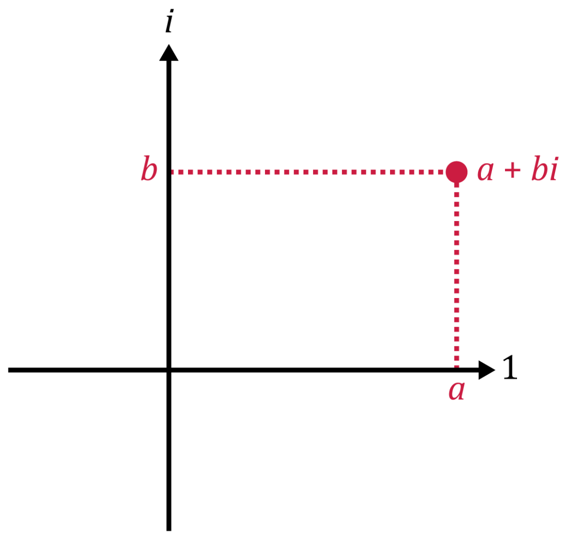


Figure 2.8: Representing a complex number as a point in the plane. Representing a complex number as a point in the plane. We draw the complex number $a + bi$ as a point with Cartesian coordinates $(a, b)$. To indicate that this is a diagram representing complex numbers we label the axes with 1 and $i$
Long description: The axes are marked in black, as with Cartesian diagrams but with the labels changed to 1 (for horizontal) and $i$ (for vertical). The point is marked in dark red, with the label $a + bi$ next to it. There is a horizontal line (light red, dashed) from the marked point to the vertical axis, where it is labelled $b$, and a vertical line from the marked point to the horizontal axis, where it is labelled $a$.

**The real number line, in the complex plane**
We often describe the real numbers $\mathbb{R}$, as being a subset of the complex numbers $\mathbb{C}$. Given any real number $a$ we can view it as the complex number $a + 0i$. We call this an **embedding** of the real numbers inside complex numbers, because it preserves all the properties of real numbers as numbers, such as addition and multiplication. (We can even identify the real number-line with the horizontal axis of $\mathbb{R}^2$.) In the rest of this module, we will simply take this embedding for granted, and allow for the possibility that a complex number could be real-valued, if it has an imaginary part of 0.

**Adding complex numbers**
If we represent two complex numbers as vectors in $\mathbb{R}^2$, then adding those two complex numbers together corresponds to adding the two vectors together, and vice-versa.

**The 'norm' of a complex number**
The **norm** of a complex number (sometimes called its 'modulus') is the natural generalisation of the concept of 'absolute value' to $\mathbb{C}$. We can easily describe this as the length

of the corresponding vector in $\mathbb{R}^2$:

$$|a + bi| := \sqrt{a^2 + b^2}\ . \tag{2.38}$$

Because we think of $z \in \mathbb{C}$ as a number rather than a vector, we use single vertical lines to denote the norm $|z|$.

### 2.2.2 Operations on complex numbers as linear transformations

Many of the operations of interest on complex numbers, can be represented in terms of linear transformations of 'the complex plane'.

**Real and imaginary components**
We can describe the real part $\mathrm{Re}(a + bi)$ and imaginary part $\mathrm{Im}(a + bi)$ of complex numbers, in terms of functions, $\mathrm{Re} : \mathbb{C} \to \mathbb{R}$ and $\mathrm{Im} : \mathbb{C} \to \mathbb{R}$. These correspond directly to the transformations $\langle 0| \,, \langle 1| : \mathbb{R}^2 \to \mathbb{R}$, which simply extract coefficients:

$$\mathrm{Re}(a+bi) \;=\; \langle 0| \cdot \begin{bmatrix} a \\ b \end{bmatrix} \qquad \text{and} \qquad \mathrm{Im}(a+bi) \;=\; \langle 1| \cdot \begin{bmatrix} a \\ b \end{bmatrix} \tag{2.39}$$

**Complex conjugation and inverses**
An operation on the complex numbers which is practically important in many applications is **complex conjugation**, which consists of negating the imaginary component of a complex number. For $z = a + bi \in \mathbb{C}$, we write

$$z^* \;=\; (a+bi)^* \;:=\; a - bi\ . \tag{2.40}$$

The complex conjugate of $z = a + bi \in \mathbb{C}$ has a useful relationship to the norm of $z$. We can represent this as a reflection through the horizontal axis of the 2D plane:

$$\begin{bmatrix} a \\ b \end{bmatrix} \mapsto \begin{bmatrix} a \\ -b \end{bmatrix}. \tag{2.41}$$

As a result of the fact that $i^2 = -1$, the product $zz^*$ can be described as a difference of squares, which is also a sum of squares:

$$zz^* \;=\; (a+bi)(a-bi) \;=\; a^2 - (bi)^2 \;=\; a^2 + b^2. \tag{2.42}$$

But this is just the same as the square of $|z|$:

$$zz^* \;=\; a^2 + b^2 \;=\; |z|^2\,. \tag{2.43}$$

If $z$ is non-zero, this can be used to compute the inverse of $z$:

$$\frac{1}{z} \;=\; \frac{z^*}{zz^*} \;=\; \frac{z^*}{|z|^2} \;=\; \frac{1}{|z|^2}\, z^*\ . \tag{2.44}$$

In particular, in the special case that $|z| = 1$, we have $z^{-1} = z^*$.

**Complex multiplication and polar coordinates**
Just using normal algebra, it's easy to describe the effect of multiplying a complex number $z = a + bi \in \mathbb{C}$ by a real number $r$: the result is:

$$r(a + bi) \;=\; (ra) + (rb)i \;. \tag{2.45}$$

We can see that this is the similar, to scalar multiplication of a vector in $\mathbb{R}^2$ by the same real value $r$:

$$r\begin{bmatrix} a \\ b \end{bmatrix} \;=\; \begin{bmatrix} ra \\ rb \end{bmatrix}. \tag{2.46}$$

We can also describe multiplication of $z$ by another complex number $w = c + di$ with $d \neq 0$, by considering the representation of the corresponding vector $\boldsymbol{w} = c\,|0\rangle + d\,|1\rangle$ in polar coordinates. If the polar coordinates of $\boldsymbol{w}$ are $(r, \theta)$, this can be carried over to write $w = r\big(\cos(\alpha)\,|0\rangle + i\sin(\alpha)\,|1\rangle\big)$. We can make this simpler, using another essential trigonometric formula which we expect you to remember:

**Proposition** (Euler's formula): $\mathrm{e}^{i\alpha} = \cos(\alpha) + i\sin(\alpha)$.

(For example: this gives us the famous equality $\mathrm{e}^{i\pi} = -1$.) We don't have the time to explain this fact, so we treat it as a definition – but in all other respects, we assume the usual rules for exponents (such as $\mathrm{e}^{ix}\,\mathrm{e}^{iy} \;=\; \mathrm{e}^{ix+iy}$). We can use this, to describe what we call the **polar decomposition of complex numbers**: if the vector $\boldsymbol{w} = c\,|0\rangle + d\,|0\rangle$ has polar coordinates $(r, \theta)$, then the complex number $w = c + di$ may be represented as

$$w \;=\; r\,\mathrm{e}^{i\alpha}\;. \tag{2.47}$$

Here, $r = |w|$ is just the norm of the complex number $w$. The angle $\alpha$ is sometimes called the **phase angle** (or the 'argument') of $w$, though we will often refer to the expression $\mathrm{e}^{i\alpha}$ as the **phase** of $w$. If we also represent $z = a + bi$ in polar form, by $z = s\,\mathrm{e}^{i\theta}$ for $s = |z|$ and some angle $\theta$, we can then multiply complex numbers by writing

$$w\,z \;=\; (r\,\mathrm{e}^{i\alpha})\,(s\,\mathrm{e}^{i\theta}) \;=\; (rs)\,\mathrm{e}^{i\alpha}\,\mathrm{e}^{i\theta} \;=\; (rs)\,\mathrm{e}^{i(\theta+\alpha)}. \tag{2.48}$$

What this describes is multiplying the norm of $z$ by a factor of $r$ and adding $\alpha$ to the phase angle of $z$. That is: multiplying $z$ by $w \in \mathbb{C}$ has the combined effect

$$z \;=\; s\,\mathrm{e}^{i\theta} \;\xmapsto{\text{multiply by } r}\; (sr)\,\mathrm{e}^{i\theta} \;\xmapsto{\text{multiply by } \mathrm{e}^{i\alpha}}\; (sr)\,\mathrm{e}^{i(\theta+\alpha)} \tag{2.49}$$

causing a (possible) change in norm, followed by a (possible) rotation of the phase angle. (This is illustrated in Figure 2.9) It is also useful to notice that – because the polar angle $\theta$ in the plane, is negated if we reflect through the horizontal axis – this also means that we can describe complex conjugation in terms of the polar form:

$$z = s\,\mathrm{e}^{i\theta} \quad\Longrightarrow\quad z^* = s\,\mathrm{e}^{-i\theta}\;. \tag{2.50}$$

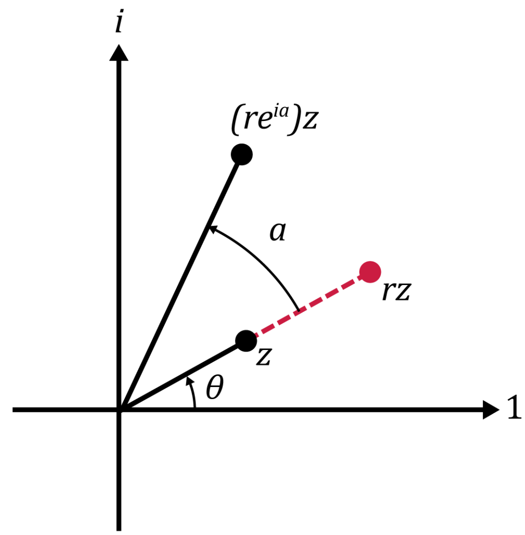


Figure 2.9: Multiplying a complex number $z$ by another complex number $re^{i\alpha}$. We draw the complex number $z$ as a point with some phase angle $\theta$. We also indicate the result of multiplying $z$ by the real number $r > 1$, which results in a complex number $rz$ with greater norm. We then represent a third complex number, which is the result of multiplying $rz$ by the complex phase $\mathrm{e}^{i\alpha}$, the effect of which is a rotation in the plane by angle $\alpha$. The resulting complex number is $(r\mathrm{e}^{i\alpha})z$.
Long description: The axes are marked in black, as with Cartesian diagrams but with the labels changed to 1 (for horizontal) and $i$ (for vertical). The complex number $z$ is indicated as a point in the upper-right quadrant. There is a black line connecting the point indicated by $z$ to the origin, and an arc from the horizontal axis to this line, labeled by a phase angle of $\theta$. This line is extended by a dashed red line, to another point labeled $rz$. A third point is indicated, still in the upper left quadrant but at a anticlockwise angle from the other two points. This point is labeled by $(r\mathrm{e}^{i\alpha})z$, and is at the same distance from the origin as $rz$. A solid line connects this point to the origin, and there is an arc to this line from the one connecting $rz$ to the origin. This arc is labeled by a rotation angle of $\alpha$.

### 2.2.3 Complex-valued vectors and matrices

We've described nearly everything that we need to know about complex numbers in themselves. Now, we will consider how we extend our ideas of inner products and linear transformations, for **vectors** of complex numbers.

We can populate vectors and matrices with complex numbers, just as we do for real numbers. We also extend the usual definitions of matrix multiplication and addition: the only difference is that the coefficients which we add and multiply together would be complex numbers, and so would be computed using the arithmetic described above. The definition of a 'linear transformation' also remains the same: the only difference is that a linear combination of vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots \in \mathbb{C}^n$, can involve complex-values coefficients, $a_1\boldsymbol{v}_1 + a_2\boldsymbol{v}_2 + \cdots + a_n\boldsymbol{v}_n$ for $a_1, a_2, \ldots \in \mathbb{C}$. We also have the same relationship between linear transformations $f : \mathbb{C}^m \to \mathbb{C}^n$, and the $m \times n$ matrices over $\mathbb{C}$.

We define the complex conjugate $M^*$ of a matrix $M$, simply by taking the complex conjugate of each of the coefficients of $M$. We use this to define an important operation on complex-valued matrices:

**Definition 1** (adjoint)**.** For any matrix $M$ with real or complex entries, including row vectors and column vectors, the adjoint of $M$ is $M^\dagger = (M^*)^\top$.

(We may also say '$M$-dagger' for the matrix $M^\dagger$.) For instance: for any column vector $\boldsymbol{v} = v_0 |0\rangle + v_1 |1\rangle + \cdots + v_{n-1} |n{-}1\rangle \in \mathbb{C}^n$, the adjoint of $\boldsymbol{v}$ is the row vector $\boldsymbol{v}^\dagger = v_0^* \langle 0| + v_1^* \langle 1| + \cdots + v_{n-1}^* \langle n{-}1|$. Because the transpose and the complex conjugate are both self-inverse operations on matrices (and independent of one another), whenever we have a relationship of the form $P = Q^\dagger$, we also have $P^\dagger = Q$ – because $P^\dagger = (Q^\dagger)^\dagger$, and transposing the coefficients of $Q$ twice and taking their complex conjugates twice is just a complicated way of doing nothing to $Q$.

We use the adjoint of vectors, to define the notion of the inner product of complex-valued vectors:

**Definition 2** (inner products)**.** For $\boldsymbol{v}, \boldsymbol{w} \in \mathbb{C}^n$, we define the inner product $\langle \boldsymbol{v}, \boldsymbol{w} \rangle = \boldsymbol{v}^\dagger \boldsymbol{w} \;=\; v_1^* w_1 + v_2^* w_2 + \cdots + v_n^* w_n$.

For unit vectors $|v\rangle$ in particular, this motivates us to define the corresponding bra-vector:

**Definition 3** (bra-vectors)**.** For $|v\rangle \in \mathbb{C}^n$ a unit vector, we define $\langle v| = |v\rangle^\dagger$.

Then, for unit vectors $|v\rangle, |w\rangle \in \mathbb{C}^n$, $\langle v | w \rangle$ is the inner product of $|v\rangle$ and $|w\rangle$. The definition of the inner product for a complex-valued vector, also motivates an extension of the definition of a norm:

**Definition 4** (norms of complex-valued vectors)**.** The norm of a vector $\boldsymbol{v} \in \mathbb{C}^n$ is

$$\big\|\boldsymbol{v}\big\| \;:=\; \sqrt{\langle \boldsymbol{v}, \boldsymbol{v} \rangle} \;=\; \sqrt{v_1^* v_1 + v_2^* v_2 + \cdots + v_n^* v_n} \;=\; \sqrt{|v_1|^2 + |v_2|^2 + \cdots + |v_n|^2}\,. \quad (2.51)$$

For real-valued vectors, these definitions coincide with the descriptions we gave earlier.

## 2.3 Eigenvectors, eigenvalues, and normal matrices

We have encountered some important operations on the 2D plane, and can describe in terms of real-valued $2 \times 2$ matrices. Describing these matrices in terms of their coefficients, however, is not always efficient or the most helpful way to describe them.

In each case, it is more helpful to emphasise how they may be described in terms of a specific choice of orthogonal unit vectors. We can do this because these are all examples of 'normal' matrices – a term which we will define below.

### 2.3.1 Reflections and rotations in 2D

#### (a) Eigenvectors and eigenvalues of a $2 \times 2$ normal matrix

A $2 \times 2$ normal matrix $M$ can be written in the form:

$$M \;=\; a_0 \left|u_0\right\rangle\!\left\langle u_0\right| \;+\; a_1 \left|u_1\right\rangle\!\left\langle u_1\right| \;. \tag{2.52}$$

Here, $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$ are two orthogonal unit vectors. As we noted in our discussion of projections in Section 2.1.3(a), this means that we can write any $\boldsymbol{v} \in \mathbb{R}^2$ (and in fact, any $\boldsymbol{v} \in \mathbb{C}^2$) as a linear combination of the form $\boldsymbol{v} = c_0 \left|u_0\right\rangle + c_1 \left|u_1\right\rangle$ (as we illustrated in Figure 2.1). The two terms:

$$\left|u_0\right\rangle \left\langle u_0\right| \,, \qquad\qquad \left|u_1\right\rangle \left\langle u_1\right| \tag{2.53}$$

are projectors onto the axes defined by these vectors. This means that we can consider the effect of the matrix $M$, not through an array of coefficients, but through how it acts on the special vectors $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$.

We call $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$ the **eigenvectors** of $M$. (This is a German word, essentially meaning 'special vector which relates to $M$' or 'vector which helps to characterise $M$'.) The way in which $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$ characterise $M$, is that $M$ has the effect of multiplying them by a scalar factor:

$$M \left|u_0\right\rangle \;=\; a_0 \left|u_0\right\rangle , \qquad\qquad M \left|u_1\right\rangle \;=\; a_1 \left|u_1\right\rangle . \tag{2.54}$$

The vectors $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$ are only 'slightly' changed by $M$: the only difference being multiplication by some (possibly complex-valued) scalar $a$. While the effect of $M$ on other vectors $\boldsymbol{v} = c_0 \left|u_0\right\rangle + c_1 \left|u_1\right\rangle$ usually wouldn't be to multiply them by a scalar, we can simply use what we know of the action of $M$ on $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$:

$$M\boldsymbol{v} \;=\; c_0 a_0 \left|u_0\right\rangle \;+\; c_1 a_1 \left|u_1\right\rangle . \tag{2.55}$$

Then, the decomposition $M \;=\; a_0 \left|u_0\right\rangle\!\left\langle u_0\right| \;+\; a_1 \left|u_1\right\rangle\!\left\langle u_1\right|$ describes how to transform any vector $\boldsymbol{v}$, by breaking $\boldsymbol{v}$ into its $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$ components, multiplying the pieces by $a_0$ or $a_1$ as appropriate, and then putting the pieces back together.

Of course, to describe how $M$ transforms vectors in Eqn. (2.55), the values of $a_0$ and $a_1$ are as important as the vectors $\left|u_0\right\rangle$ and $\left|u_1\right\rangle$. The specific values of $a_0$ and $a_1$ associated with a matrix $M$, are called its **eigenvalues**. (This is a German word again: it means 'scalar value which helps to characterise $M$'.) We may describe the eigenvectors $\left|u_0\right\rangle$ as being an **$a_0$-eigenvector of $M$**, and $\left|u_1\right\rangle$ as being an **$a_1$-eigenvector of $M$**. Through Eqn. (2.52), we see that the eigenvalues and the eigenvectors together that characterise the matrix: like a decomposition of a vector into an orthogonal basis, but for matrices.

If you are simply given a $2 \times 2$ matrix $M$, you can't always tell what the eigenvectors and eigenvalues of $M$ will be by inspecting the coefficients. There are procedures to determine the eigenvectors and eigenvalues, but in this module, we will either tell you the eigenvectors and eigenvalues of a matrix, or give you enough information to find them easily.

Not all $2 \times 2$ matrices can be decomposed as in Eqn. (2.52): this decomposition is special to 'normal' matrices (which we will define further below). Eigenvectors and eigenvalues can still be used to characterise some matrices which are not 'normal', but it is slightly more complicated: see the Supplemental Note on 'Eigen-decompositions and non-normal matrices' for more information. For us, it will be enough to consider matrices which can be decomposed as in Eqn. (2.52), and its generalisation to $n \times n$ matrices for $n \geqslant 2$.

### (b) Eigenvalues in relation to 2D rotations, reflections, and projections

When it comes to the question of whether a matrix $M$ is a reflection, a rotation, or a projection, this is easy to see if we are provided with a description of $M$ by its 'eigen-decomposition' – its decomposition as in Eqn. (2.52). For $2 \times 2$ matrices over $\mathbb{R}$, the eigenvectors and eigenvalues will sometimes be clearly related to what we saw in Section 2.1.3; but even when this is not the case, the eigen-decomposition provides useful information about what the matrix does.

**Projections**
We have already seen how an orthogonal projection can be represented in a form very like Eqn. (2.52): for an orthogonal projector $P$ onto the axis defined by a vector $|u_0\rangle$, if $|u_1\rangle$ is an orthogonal unit vector to $|u_0\rangle$, we have

$$P \;=\; 1 \cdot |u_0\rangle\langle u_0| + 0 \cdot |u_1\rangle\langle u_1| \;. \tag{2.56}$$

Then, $|u_0\rangle$ is a $+1$-eigenvector of $P$, and $|u_1\rangle$ is a 0-eigenvector of $P$. (In this case, we would more often say that $|u_1\rangle$ is in the **nullspace** of $P$, though describing it as a 0-eigenvector is also correct.)

**Reflections**
As we also saw above, we can describe a reflection $R$ in terms of a pair of orthogonal unit vectors $|v_0\rangle$ and $|v_1\rangle$, which maps a vector $c_0\,|v_0\rangle + c_1\,|v_1\rangle$ to the vector $c_0\,|v_0\rangle - c_1\,|v_1\rangle$. We can describe this by the eigendecomposition

$$R \;=\; |v_0\rangle\langle v_0| - |v_1\rangle\langle v_1|\,, \tag{2.57}$$

so that $|v_0\rangle$ is again a $+1$-eigenvector of $R$, and $|v_1\rangle$ is a $-1$-eigenvector of $R$.

**Rotations**
Notice that almost all $2 \times 2$ rotations $Q$ don't preserve or re-scale any vectors at all in the 2D plane, apart from $\mathbf{0}$. The two exceptions are the rotation by 0 radians (which is $I$) and the rotation by $\pi$ radians (which is $-I$), which either preserve or negate **all** vectors in $\mathbb{R}^2$. In fact, however, all $2 \times 2$ real-valued rotations have a pair of orthogonal eigenvectors – albeit, ones with complex-valued coefficients. Let

$$|w_0\rangle \;=\; \tfrac{1}{\sqrt{2}}\,|0\rangle + \tfrac{i}{\sqrt{2}}\,|1\rangle\;, \qquad\qquad |w_1\rangle \;=\; \tfrac{1}{\sqrt{2}}\,|0\rangle - \tfrac{i}{\sqrt{2}}\,|1\rangle\;. \tag{2.58}$$

Then a rotation $Q$ of the 2D plane by an angle $\alpha$, has an eigendecomposition of the form

$$Q \;=\; \mathrm{e}^{-i\alpha}\,|w_0\rangle\langle w_0| + \mathrm{e}^{i\alpha}\,|w_1\rangle\langle w_1| \;. \tag{2.59}$$

This may look deeply mysterious – and, in fairness, there is no straightforward 2D geometric interpretation of the fact that $|w_0\rangle$ and $|w_1\rangle$ are eigenvectors of every rotation operation. But we can actually verify that Eqn. (2.59) describes a real-valued matrix describing a rotation by $\alpha$, using nothing but algebra, Euler's identity, and the facts that $\cos(-\alpha) = \cos(\alpha)$ and $\sin(-\alpha) = -\sin(\alpha)$:

$$\begin{aligned} Q \;&=\; \mathrm{e}^{-i\alpha}\,|w_0\rangle\langle w_0| + \mathrm{e}^{i\alpha}\,|w_1\rangle\langle w_1| \\ &=\; \Bigl(\frac{\cos(-\alpha) + i\sin(-\alpha)}{2}\Bigr)\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} + \Bigl(\frac{\cos(\alpha) + i\sin(\alpha)}{2}\Bigr)\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \\ &=\; \Bigl(\frac{\cos(\alpha) - i\sin(\alpha)}{2}\Bigr)\begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} + \Bigl(\frac{\cos(\alpha) + i\sin(\alpha)}{2}\Bigr)\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \\ &=\; \begin{bmatrix} \cos(\alpha) & i^2\sin(\alpha) \\ -i^2\sin(\alpha) & \cos(\alpha) \end{bmatrix} \;=\; \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} \;. \end{aligned} \tag{2.60}$$

You are not expected to memorise Eqn. (2.59), or to be able to form any intuitions about it: we demonstrate this equality in order to demonstrate that even rotation matrices can be described in terms of eigenvectors and eigenvalues (and, to show how complex numbers can be important even for a $2 \times 2$ real-valued matrix).

### 2.3.2 Basics of normal matrices

In this module, we will be interested almost exclusively in 'normal' matrices, which we are now in a position to define:

**Definition 5.** An $n \times n$ (real or complex) matrix $M$ is normal, if it has an eigendecomposition $M = a_0\,|v_0\rangle\,\langle v_0| + a_1\,|v_1\rangle\,\langle v_1| + \cdots + a_{n-1}\,|v_{n-1}\rangle\,\langle v_{n-1}|$, for some eigenvalues $a_0, a_1, \ldots, a_{n-1} \in \mathbb{C}$ and some orthogonal unit vectors $|v_0\rangle\,, |v_1\rangle\,, \ldots, |v_{n-1}\rangle \in \mathbb{C}^n$.

This is not quite how a 'normal' matrix is usually defined, but it is closely related to the usual definition: and it describes what for us is the most important feature of a normal matrix – the fact that it can be characterised by an orthonormal basis of eigenvectors.

In fact, we are almost exclusively interested in two specific kinds of 'normal' matrices, which are distinct but closely related.

**Definition 6.** An $n \times n$ (real or complex) matrix $M$ is **Hermitian**, if it is normal and has only real eigenvalues: $a_0, a_1, \ldots, a_{n-1} \in \mathbb{R}$.

**Definition 7.** An $n \times n$ (real or complex) matrix $M$ is **unitary**, if it is normal and has only eigenvalues which are complex phases, $a_k = \mathrm{e}^{i\theta_k}$ for various angles $\theta_k \in \mathbb{R}$.

These two kinds of normal matrix play an important role in quantum mechanics, and in quantum computation in particular. We now discuss these two kinds of operator in overview, with the aim of providing some intuitions of how to think about them, which hopefully will prove useful throughout the module.

### (a) Adjoints and eigendecompositions

Before proceeding, it will be helpful to talk about the adjoints of normal matrices, and how they interact with the eigenvalues and eigenvectors of a normal matrix.

Recall that because $(AB)^\top = B^\top A^\top$, it follows also that $(AB)^\dagger = B^\dagger A^\dagger$. Then for any unit vector $|v\rangle \in \mathbb{C}^n$, we have $\bigl(\,|v\rangle\langle v|\,\bigr)^\dagger \;=\; \langle v|^\dagger\, |v\rangle^\dagger \;=\; |v\rangle\langle v|$. That is, all orthogonal projectors are **self-adjoint**: they are equal to their own adjoints. This has consequences for normal matrices in general. Consider a normal matrix

$$M \;=\; \sum_{j=0}^{n-1} a_j\, |v_j\rangle\, \langle v_j| \;\;, \tag{2.61}$$

with eigenvectors $|v_j\rangle$ and eigenvalues $a_j$. Because the complex conjugate and the transpose both distribute over sums, the adjoint of $M$ is

$$M^\dagger \;=\; \left(\sum_{j=0}^{n-1} a_j\, |v_j\rangle\langle v_j|\right)^\dagger \;=\; \sum_{j=0}^{n-1}\Bigl(a_j\, |v_j\rangle\langle v_j|\Bigr)^\dagger \;=\; \sum_{j=0}^{n-1} a_j^*\, |v_j\rangle\langle v_j| \;\;. \tag{2.62}$$

What this means is that $M^\dagger$ has exactly the same eigenvectors as $M$ does, and is different only in its eigenvalues. Specifically: the eigenvalues of $M^\dagger$ are each the complex conjugate of the corresponding eigenvalues of $M$.

### (b) Hermitian matrices

As we note above: a **Hermitian** matrix is a normal matrix (having $n$ orthogonal eigenvectors) and only eigenvalues over $\mathbb{R}$. This abstract description gives us a useful way to recognise them, which is actually the more common way to define them:

**Proposition 2** (Hermitian operators are self-adjoint)**.** *An $n \times n$ matrix $M$ is Hermitian, if and only if $M = M^\dagger$: that is, if $M$ is self-adjoint.*

This is something that perhaps you can see easily. Consider a Hermitian operator $M$, as in Eqn. (2.61): because the eigenvalues $a_j$ are all real-valued, we have $a_j = a_j^*$, so that

$$M^\dagger \;=\; \sum_{j=0}^{n-1} a_j^*\; |v_j\rangle\langle v_j| \;=\; \sum_{j=0}^{n-1} a_j\, |v_j\rangle\langle v_j| \;=\; M. \tag{2.63}$$

In the reverse direction, we can also see that if $M$ is a normal matrix which is self-adjoint, all of its eigenvalues $a_j$ must be real-valued: For each eigenvector $|v_j\rangle$, we would have

$$a_j\,|v_j\rangle \;=\; M\,|v_j\rangle \;=\; M^\dagger\,|v_j\rangle \;=\; a_j^*\,|v_j\rangle \;, \tag{2.64}$$

so that $a_j = a_j^*$, meaning that $a_j \in \mathbb{R}$. This provides us with a helpful way to determine that a matrix is Hermitian. But for us, the more important property of a Hermitian matrix is that all of its eigenvalues are real.

**The purpose Hermitian matrices**

Hermitian matrices will be important, in that they are an indirect but essential way of describing measurable properties, that could hold for a computation or a physical system. We can describe this in outline, as follows:

- **Projectors are Hermitian operators** – because their eigenvalues are all 0 or 1, those eigenvalues are all real numbers. We can think of them as operators, which select for the components of a vector $\boldsymbol{v}$ which satisfy some particular property (by 'annihilating' the orthogonal components of $\boldsymbol{v}$ which do not satisfy that property). We already saw this in Week 1, when considering conditional probabilities, in the special case of the projectors $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ on individual bits.

- **Reflections are Hermitian operators** – because their eigenvalues are all $\pm 1$, those eigenvalues are again all real-valued. Some times, rather than selecting for a component of a vector $\boldsymbol{v}$ which satisfies some property, it is more helpful to multiply the component for which that property holds by $-1$: if we do this for more than one property, we can combine those results as a product of signs.

- **All 'measurable' physical properties are represented by Hermitian operators** – in this module, it will **not** be important to understand quantum mechanics as such. But in quantum mechanics, all measurements are in effect done by evaluating some property of a quantum wave function, with respect to an operator $H$ which represents a physical quantity (for example, energy). The outcome of an individual measurement is then some eigenvalue of the operator: though the specific outcome is generally a randomly selected eigenvalue, according to a specific probability distribution.

Some of the above ideas will be made more precise, later in the module: we will return to these ideas when needed.

### (c) Unitary matrices

As we define them above, a unitary matrix is a normal matrix whose eigenvalues are all complex phases: numbers of the form $\mathrm{e}^{i\theta}$ for real-valued angles $\theta \in \mathbb{R}$. This definition generalises both the notion of a rotation and a reflection: every rotation is a unitary transformation, and every reflection is also a unitary transformation.

As with Hermitian matrices, this definition of unitary matrices can be connected to a direct way of recognising them, which is often used as a definition:

**Proposition 3** (The inverses of unitary matrices are their adjoints)**.** *An $n \times n$ matrix $U$ is unitary, if and only if $U$ is invertible and $U^{-1} = U^\dagger$; equivalently, if and only if $U^\dagger U = UU^\dagger = I$.*

How do we know this is true? Consider a normal matrix $U$, and in particular what can be said about the product $UU^\dagger$:

$$\begin{aligned} UU^\dagger \;&=\; \left(\sum_{j=0}^{n-1} a_j\,|v_j\rangle\langle v_j|\right)\left(\sum_{k=0}^{n-1} a_k\,|v_k\rangle\langle v_k|\right)^{\!\dagger} \;=\; \left(\sum_{j=0}^{n-1} a_j\,|v_j\rangle\langle v_j|\right)\left(\sum_{k=0}^{n-1} a_k^*\,|v_k\rangle\langle v_k|\right) \\ &=\; \sum_{j=0}^{n-1}\sum_{k=0}^{n-1} a_j\,a_k^*\;|v_j\rangle\,\langle v_j\,|v_k\rangle\,\langle v_k| \;\;=\;\; \sum_{j=0}^{n-1} a_j\,a_j^*\;|v_j\rangle\langle v_j|\;\;. \end{aligned} \tag{2.65}$$

In the last equality, we make use of the fact that $\langle v_j\,|v_k\rangle = 1$ if $j = k$, and that $\langle v_j\,|v_k\rangle = 0$ otherwise. (This is because the vectors $|v_j\rangle$ are an orthogonal basis of unit vectors.) This allows us to eliminate the sum over $k$ except for the terms where $k = j$ .

Thus, $UU^\dagger$ is a matrix with the same eigenvectors as $U$ and $U^\dagger$, and eigenvalues $a_j a_j^* = |a_j|^2$. (We can similarly show that $U^\dagger U$ evaluates to the same operator.) If $U$ is unitary, then we have $|a_j| = 1$ for every $j$. This means that $a_j^* = a_j^{-1}$ for each $j$, so that

$$UU^\dagger \;=\; \sum_{j=0}^{n-1}\,|v_j\rangle\langle v_j|\;\;. \tag{2.66}$$

The right-hand side of this equation is what is known as a **resolution of the identity**: that is, a way of representing the identity matrix $I$, as a sum of orthogonal projectors. (To see that this is the identity, consider how this matrix acts on a vector $\boldsymbol{v} = c_0\,|v_0\rangle + c_1\,|v_1\rangle + \cdots + c_{n-1}\,|v_{n-1}\rangle$.) So if $U$ is unitary, then $UU^\dagger = U^\dagger U = I$. In the reverse direction, if $UU^\dagger = I$, this means that each of the eigenvalues $a_j a_j^*$ of $UU^\dagger$, satisfies $|a_j|^2 = 1$. Then, every eigenvalue $a_j$ of $U$ is a phase factor $\mathrm{e}^{i\theta_j}$ for some $\theta_j \in \mathbb{R}$.

**The purpose of unitary matrices**

Unitary matrices will be important, in that they will describe the way in which we can transform information to realise computations. We can describe this in outline, as follows.

- **Rotations and reflections are unitary.** When describing how to do quantum computation at its lowest level, we will soon see how rotations play an important role in describing how quantum data can be transformed. Reflections will also prove to be important in understanding particular quantum algorithms.

- **Permutation operations are unitary.** Recall that a permutation matrix $P$ is one which has exactly one coefficient 1 in each row and in each column, with zeroes everywhere else. We noted in Week 1 that this means that $P^{-1} = P^\top$. Because $P$ is a real-valued matrix we in fact have $P^\top = (P^*)^\top = P^\dagger$ Then, we also have $P^{-1} = P^\dagger$, or in other words, $PP^\dagger = I$.

  Permutation matrices are one of the most important sort of unitary transformation in quantum algorithms: the fact that they are normal matrices with eigenvalues which are all complex phases is important in some of the algorithms we consider.

The property $UU^\dagger = U^\dagger U = I$ is an important property, which determines the role of unitary matrices in quantum computation. But it is the property that the eigenvalues of $U$ are all complex phases that will be essential to understand quantum **algorithms**.

## 2.4 Single-qubit states and measurements

We have now considered nearly enough preliminary concepts, that we may begin to consider the subject of quantum information. Specifically, we can begin to consider what sort of information can be stored by a single **qubit** (which is a 'quantum bit').

We begin with the very simplest aspects of qubits: what sort of information they can store, how we can get that information out again, and the way that we represent these concepts using vectors and matrices.

### 2.4.1 Preamble: directions and rotations in 3D space

The way that data can be stored in a single qubit, is closely connected with the geometry of the 'unit sphere' in 3D: the set of points at distance 1, in any direction, in $\mathbb{R}^3$. We now consider how to describe this, as a preliminary to understanding qubits.

You can think of the unit sphere as being similar to the points on a globe – and we can describe directions in 3D space in a similar way. This is illustrated in Figures 10 and 11, which show how we can describe points on the unit sphere in terms of an **azimuthal angle** (analogous to longitude) and a **polar angle** (which is similar to latitude, but measured from the north pole instead of the equator).

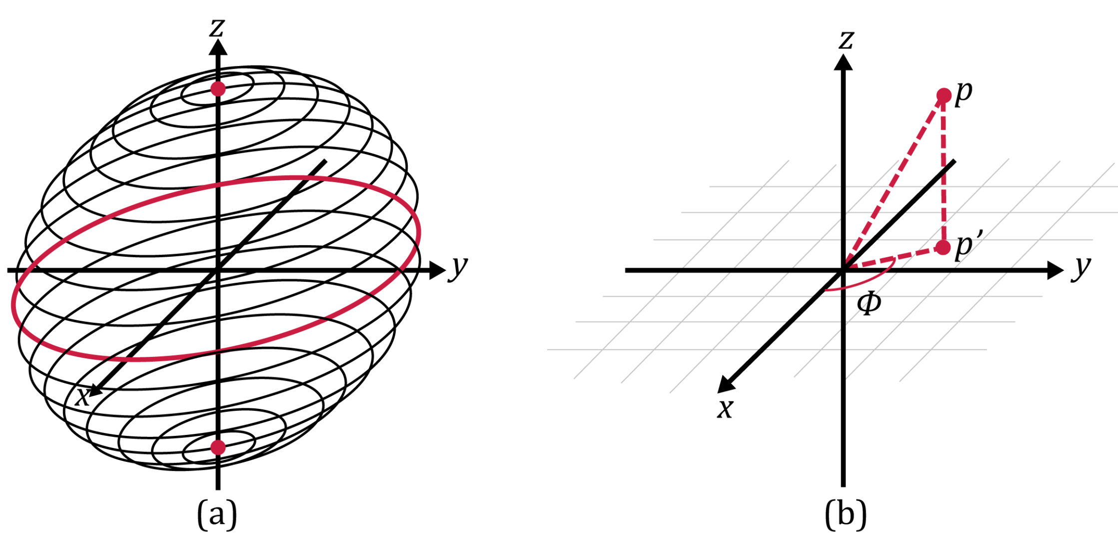

Figure 2.10: The unit sphere and the notion of direction in 3D space. The unit sphere in 3D is just the set of all points at a distance of 1 from the origin $(0, 0, 0)$ defined by the $x$, $y$, and $z$ axis. We think of $z$ as the vertical axis, and the $x$ and $y$ axes as being horizontal. Here, the $x$ axis is drawn pointing out of the page, and to the left.

**(a)** The locations on the unit sphere can be thought of as analogous to points on the globe, with a 'north pole' at the top on the positive $z$-axis, a 'south pole' at the bottom on the negative $z$-axis, and an 'equator' where the sphere meets the $x$-$y$ plane. The circles on the globe which are parallel to the equator can be thought of as lines of constant 'latitude', which are each at some angle 'north' or 'south' of the equator. (For us it will be more helpful to describe these in terms of **polar angles**, each at some fixed angle from the north pole.) Within each of these circles, we can consider an angle of 'longitude' that each point lies at, away from the half-plane described by the $z$ axis and the positive $x$ axis.

**(b)** The 'azimuthal' angle $\phi$ of a point $\boldsymbol{p}$ in 3D space. For a point on the unit sphere, the angle $\phi$ corresponds to an angle of longitude, measured 'east' of the positive $x$ axis (in other words, anticlockwise in the $x$-$y$ plane). This can be described in terms of the projection (or 'shadow') $\boldsymbol{p}'$, which is cast by the point $\boldsymbol{p}$ onto the $x$-$y$ plane. The $x$-$y$ plane itself is indicated by a light gray grid pattern.
Long description: Two diagrams in Cavalier projection, labeled **(a)** and **(b)**. In each, the $x$, $y$, and $z$ axes are marked in black. The $x$ axis points down and to the left, the $y$ axis points right, and the $z$ axis points up. In the diagram **(a)**, a sphere is illustrated as a vertical array of parallel circles, aligned with the $x$ and $y$ axes. The north pole (top of the sphere) is marked in dark red, as is the equator, while the south pole (bottom of the sphere) is marked in light red. In the diagram **(b)**, the unit sphere is not indicated, but two points $\boldsymbol{p}$ and $\boldsymbol{p}'$ are. The point $\boldsymbol{p}$ lies above the $x$-$y$ plane, which has been cross-hatched in a light grey. The point $\boldsymbol{p}'$ lies within the $x$-$y$ plane, directly below $\boldsymbol{p}$. The lines between $\boldsymbol{p}$, $\boldsymbol{p}'$, and the origin are indicated by dashed lines. The azimuthal angle $\phi$, between the $x$ axis and the line connecting the origin to $\boldsymbol{p}'$, is marked in dark red.

**The polar angle** $\theta$, describes how close a point on the unit sphere is to the north pole – or, how close a direction in 3D is to being aligned with the positive $z$ axis. We may take $0 \leqslant \theta \leqslant \pi$ in this case: we don't need to indicate positive or negative angles, or any angle greater than $\pi$. A polar angle $\theta = 0$ indicates alignment with the $z$-axis (straight up), $\theta = \pi$ indicates pointing in the opposite direction as the $z$-axis (straight down). In between, we have $\theta = \pi/2$, which describes a direction within the $x$-$y$ plane. But in order to describe which direction **within** this plane, we need more information.

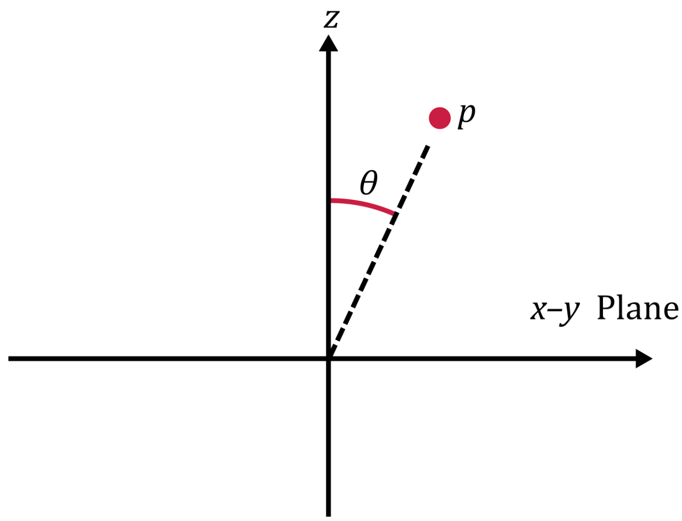


Figure 2.11: The polar angle. For a point $\boldsymbol{p}$ in 3D space, the 'polar angle' of $\boldsymbol{p}$ is the angle between the $z$ axis, and the line between $\boldsymbol{p}$ and the origin. (In this case, we do not need to consider $\theta$ as having a direction.) Long description: A diagram in orthogonal projection. The vertical axis is marked $z$, and the horizontal axis is marked '$x$-$y$ plane', as though viewing that plane edge-on. The point $p$ has been marked in light red, and a dashed line goes from the origin to the point $p$. The polar angle $\theta$ is marked in dark red, tracing the angle from the $z$ axis down to the dashed line between the origin and $p$.

**The azimuthal angle** $\phi$, can be thought of as describing a direction in or along the $x$-$y$ plane, where we take the projection (or shadow) of a point $\boldsymbol{p}$ onto the $x$-$y$ plane. This corresponds to the usual way you would consider an angular position in 2D space, described in terms of a anticlockwise angle from the positive $x$ axis. (If the polar angle is either 0 or $\pi$, there is no well-defined azimuthal angle, in the same way that the north pole and the south pole don't have any longitude east or west.)

The two angles $\theta$ and $\phi$ uniquely determine a direction in 3D space. (To describe a specific point in 3D space, we can also specify a distance from the origin, or 0 in the case of the origin itself: though for us this will not be important.)

### 2.4.2 High-level physical motivation for qubits

To describe the sort of information that a single qubit can store, we consider some ideas from physics in broad outline. You do not need to study these ideas in any detail: we present them here just as motivation to support the mathematical definitions to follow.

#### (a) Atoms, magnetic fields and the Stern–Gerlach experiment

Silver atoms respond to magnetic fields. The property that determines what impact a magnetic field has on an a single atom of silver, is called the 'spin' of the atom. This property is not specific to silver atoms: other atoms, ions, and particles also have spin.

You do not need to understand spin in any detail in this module. If you like, you can think of it (very naïvely!) as being similar to a slim bar magnet which is pointing in a particular direction, spinning around its axis. A bar magnet would react to a magnetic field, just as a compass responds to the magnetic field of the Earth. We can characterise the way that the bar magnet would respond in terms of the rate at which the magnet is spinning (the 'amount' of spin), and the direction that the bar magnet is pointing. For particles such as silver atoms or electrons, the 'amount' of spin is fixed, and the direction of the spin is just a direction in 3D space. The amount of information which is 'represented' by the bar magnet can then be described by that direction, which we can describe by a point on the unit sphere.

**However**, there is an important difference between the property of 'spin' that relates to silver atoms and electrons, and the behaviour of a bar magnet. In the case of silver atoms in particular, this was demonstrated through the 'Stern–Gerlach experiment'.

The Stern–Gerlach experiment involves firing a silver atom through a 'cavity' (a semi-enclosed space), which has a magnetic field pointing roughly in some direction (e.g., along the $z$-axis), but which decreases in intensity in that same direction. Through the physics of spinning objects and magnetism, an actual miniature spinning bar magnet would be deflected in the direction of the magnetic field, by an amount depending on how its spinning aligns with the magnetic field. (If the magnetic field points along the $z$-axis, a bar magnet which is perfectly aligned in one direction or the other with the $z$-axis would be deflected in the direction of its spin, by the greatest possible extent. To contrast, a bar magnet aligned with the $x$-axis or $y$-axis would not be deflected at all, and a bar magnet pointing somewhere between would be partially deflected.) But this is not what is observed in the Stern–Gerlach experiment. Instead, the silver atom is deflected either as much as possible in the direction of the magnetic field, or against it – with a **probability** that depends on the direction of the spin. Furthermore, the deflected atoms afterwards are perfectly aligned or opposed to the direction of the magnetic field, so that attempts to obtain more information about the original direction of the spin of the atoms don't provide any further information.

**Stern-Gerlach experiment (Figure 2.11)**
There are variations on this experiment in which you would get exactly three or more possible outcomes, but it is this two-outcome version of the Stern–Gerlach experiment which will motivate how we think of quantum information.

### (b) What a computer scientist learns from the Stern–Gerlach experiment

We will not consider the actual physics of this experiment (as this is not a physics module). But we can gather the following information about the 'spin' of certain particles such as silver atoms:

- The range of possibilities for the 'spin' is described by the points on the unit sphere.

- It is possible to perform a measurement, in any direction, to try to distinguish whether the spin points more to one of two opposite directions.
- Though the possible directions for spin are continuous, the outcome of such a measurement only yields one of two possible outcomes – the outcomes which correspond to the two opposite directions. The probability of obtaining one outcome or the other is random, but depends on the direction of the spin before measurement.
- After the measurement, the direction of the spin is changed to be consistent with the outcome that was actually realised.

Acting as computer scientists, we simply accept that this is how 'spin' can act, and consider the implications for how we can use this to store information. We have a physical system, which can be measured in various ways, but which in each case yields one of only two distinguishable outcomes; it can retrieve only a single bit of information.

#### (c) The Qubit as an abstraction

Classical computers use **bits** (0s and 1s) as the basis for computation. Precisely how they do this matters to the people who build computers, but for the most part doesn't matter to the people who use them or design algorithms for them – once we establish the ability to store two distinguishable values (that we label by '0' or '1'), we think just in terms of those two values.

For quantum computing, we instead use **qubits** (quantum bits) as the basis for computation. While the notion of 'spin' that we have described serves as a model for what can be stored by a qubit, other physical properties (such as the polarisation of light) have similar descriptions, even if the underling physics is different. For this reason – though it is important to think about such things when contributing to building quantum computers – we won't worry about the physical reasons or meaning of what we do to a qubit.

In the following, a 'qubit' is an abstraction of a physical system. It is similar to a single-bit register, but can store data in a more sophisticated way. We refer to the 'state' of a qubit (rather than its 'value') to keep in mind that we're describing the condition of **some** physical system, even if we are not committing to just what it is. We may then consider the (abstract descriptions of the) states that a qubit can have.

### 2.4.3 Representing the state of a single qubit

We take a qubit to be something whose state can be described by a point on the unit sphere, that we can only learn about through measurements which produce sharp, probabilistic outcomes representing binary distinctions.

### (a) The standard basis states

If we focus first on the possibility of only having two outcomes, this suggests that we represent a qubit by a two-dimensional vector, with the standard basis

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \text{and} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{2.67}$$

representing two opposing directions. By convention, we identify these with two opposing directions along the $z$-axis – the north pole of the unit sphere for $|0\rangle$, and the south pole for $|1\rangle$. We call these 'state-vectors' of the qubit. We sometimes refer to a vector such as $|0\rangle$ simply as a 'state', rather than a 'state-vector'. However, different vectors can describe the same state: as we will see, the vectors

$$-|0\rangle = \begin{bmatrix} -1 \\ 0 \end{bmatrix} \qquad \text{and} \qquad -|1\rangle = \begin{bmatrix} 0 \\ -1 \end{bmatrix} \tag{2.68}$$

represent the same single-qubit states as the vectors $|0\rangle$ and $|1\rangle$ respectively, corresponding to the north and south poles of the unit sphere (we generalise this below). So while we may refer, for example, to 'the state $|0\rangle$', it is important to remember that the states are only **represented** by the vectors, and are not identical to them.

Setting aside imperfections in our devices, the states $|0\rangle$ and $|1\rangle$ are perfectly distinguishable. If we perform a measurement (such as the one in the Stern–Gerlach experiment), the state $|0\rangle$ always gives one of the two possible outcomes (e.g., an upward deflection), and the state $|1\rangle$ always gives the opposite outcome (e.g., a downward deflection). While the outcome may not be certain for other states of the qubit, the outcome **is** certain for these states, which we think of as being given as a single bit: 0 or 1.

You may wonder why we represent states of a qubit, which correspond to points on a sphere in 3D, by a **two**-dimensional vector. In short, it is a consequence of the fact that we want to represent that any measurement on a qubit can only have two perfectly distinguishable outcomes, similarly to a probability vector $\boldsymbol{p} \in \mathbb{R}^2$ on a single bit. We will soon observe how single-qubit states **differ** from probability vectors: e.g., to represent points on the 3D sphere we can't just use vectors in $\mathbb{R}^2$.

### (b) The Bloch sphere

When we use it to describe the states of a single qubit, we refer to the unit sphere as **the Bloch sphere**, to emphasise the fact that it's a description of single-qubit states rather than a physical sphere. But it also emphasises the fact that it is a sphere – the most symmetrical 3D shape that there is. Every pair of angles $(\theta, \phi)$, for $0 \leqslant \theta \leqslant \pi$ and $-\pi < \phi \leqslant \pi$, represents **both** a state of a qubit, **and** a direction in which we could measure to identify that state, distinguishing it from the state directly opposite that point on the sphere represented by the angles $(\pi - \theta,\, \phi \pm \pi)$. We illustrate the Bloch sphere in Figure 2.12.

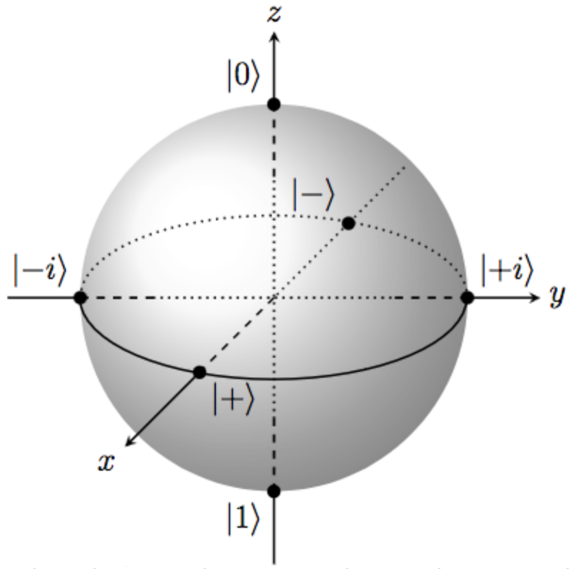


Figure 2.12: The Bloch sphere. The unit sphere, shown with some indications of where the $x$-, $y$-, and $z$-axes intersect it. The points of intersection correspond to single qubit states which have standard names: these are defined shortly below. (The states $|\pm\rangle$ and $|\pm i\rangle$ in particular are defined in Eqns. (2.72) and (2.75).)
Long description: A shaded gray ball, intersected by three lines indicating the $x$-, $y$-, and $z$-axes. The $x$-axis runs from the upper right to the lower left, using the convention which attempts to convey the direction out of the page. The $y$-axis runs horizontally across the page, and the $z$-axis runs vertically up the page. These axes intersect at the centre of the sphere. Where these axes are meant to be in the interior of the sphere, the depth at which they intersect it is indicated by representing these axes by dotted or dashed lines. There are dots Where the axes meet the surface of the sphere. The dots on the $x$-axis are labeled by $|+\rangle$ (towards the 'front' of the sphere at the lower left) and $|-\rangle$ (at the 'back' of the sphere at the upper right). The dots on the $y$-axis are labeled by $|+i\rangle$ (on the right) and $|-i\rangle$ (on the left). The dots on the $z$-axis are labeled by $|0\rangle$ (at the top) and $|1\rangle$ (at the bottom).

It is (very!) common to emphasise the $|0\rangle$ and $|1\rangle$ states as a way to store information in a qubit and, as states, to try to distinguish by measurement. But we could just as well measure along the $x$-axis, the $y$-axis, or any other axis, and obtain **different** information from the qubit. The reasons why you might do this would depend on what sort of information you were storing in the qubits, and what you were trying to do with them.

But in principle, even though we have given the labels '$|0\rangle$' and '$|1\rangle$' to the states on the top and the bottom of the Bloch sphere, that does not mean that these states are the only way in which we can use two distinguishable states to store one bit of information.

The way that we represent single-qubit states by vectors in $\mathbb{C}^2$ will reflect this symmetry in the following ways: just as the states $|0\rangle$ and $|1\rangle$ are unit vectors that are orthogonal to one another, **(a)** every unit vector $|u\rangle \in \mathbb{C}^2$ represents a single-qubit state, and every single-qubit state can be represented by a unit vector, and **(b)** a vector $|v\rangle \in \mathbb{C}^2$ represents the state on the opposite side of the Bloch sphere from $|u\rangle$, if and only if the vectors themselves are orthogonal, so that $\langle u\,|v\rangle = 0$.

**Aside.** Apart from quantum theory, there are physical reasons why single-qubit states should be unit vectors: for instance, relationships to field intensity when considering light polarisation. In our case we will simply describe this as relating to the fact that points on the unit sphere have distance 1 from the origin (though this relationship is not direct, because the unit sphere is not a sphere in $\mathbb{C}^2$).

### (c) Single qubit state-vectors and measurements in a nutshell

We are now about as ready as we will ever be, to describe how we represent single-qubit states and measurements, explicitly in terms of vectors in $\mathbb{C}^2$.

**Definition 8** (Single-qubit state-vectors)**.** A single-qubit state vector, is a unit vector

$$|\psi\rangle \;=\; a\,|0\rangle + b\,|1\rangle \;=\; \begin{bmatrix} a \\ b \end{bmatrix} \in \mathbb{C}^2 \tag{2.69}$$

– that is, for which $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$.

**Note:** The greek letter '$\psi$' (pronounced either 'psai' or 'psee') is very often used to describe state-vectors of one or more qubits. But we may also sometimes use other greek letters such as '$\phi$' or '$\alpha$', or roman letters such as '$u$' or '$v$'.

While there is more than one sort of measurement that we can perform on a single qubit, in practice we consider measurement along the $z$-axis most often, which determines how closely aligned a state $|\psi\rangle$ is with the $|0\rangle$ or $|1\rangle$ states.

**Definition 9** (Standard basis measurement)**.** A standard basis measurement (or computational basis measurement) on a single qubit, is a process which takes a qubit in some state $|\psi\rangle = a\,|0\rangle + b\,|1\rangle \in \mathbb{C}^2$ as input, and produces a single bit $r \in \{0,1\}$ as output according to the probability distribution

$$\Pr\bigl[r = 0\bigr] \;=\; |a|^2, \qquad\qquad \Pr\bigl[r = 1\bigr] \;=\; |b|^2. \tag{2.70}$$

After this operation, the qubit itself is in the state $|0\rangle$ if $r = 0$, and the state $|1\rangle$ if $r = 1$.

Note in particular that the probabilities add to 1 precisely because $|\psi\rangle$ is a unit vector. The qubit state afterwards 'updates' to $|r\rangle$ after the measurement, for $r \in \{0,1\}$ : this is referred to as **state collapse**.

### (d) Single-qubit states, measurements, and the Bloch sphere

We will now describe how the definition above relates to the Bloch sphere, to try to provide a bit more insight of how to picture single-qubit states.

**Angles on the Bloch sphere, and between state-vectors**
Two states which are described by opposite points on the Bloch sphere (separated by an

angle of $\pi$), will be described by orthogonal vectors $|u\rangle, |v\rangle \in \mathbb{C}^2$ such that $\langle u|v\rangle = 0$ – which are separated by an angle of $\pi/2$ in complex space (for example, consider the state-vectors $|0\rangle$ and $|1\rangle$, which can be pictured as vectors in $\mathbb{R}^2$). This generalises to all angular separations: states which are separated by an angle of $\theta$ on the Bloch sphere are described by state-vectors $|u\rangle, |v\rangle \in \mathbb{C}^2$, which are separated by an angle of only $\theta/2$ in complex space.

**States in the $x$-$z$ plane**
The above observation about angles is easiest to see for states in the Bloch sphere that lie on the $x$-$z$ plane, having angular coordinates $(\theta, 0)$ or $(\theta, \pi)$ for any value $0 \leqslant \theta \leqslant \pi$. For $\theta = 0$ we have the state $|0\rangle$ at the north pole, and for $\theta = \pi$ we have the state $|1\rangle$ at the south pole. In between them, there is a whole spectrum of state-vectors

$$\cos\left(\tfrac{\theta}{2}\right)|0\rangle + \sin\left(\tfrac{\theta}{2}\right)|1\rangle = \begin{bmatrix}\cos(\theta/2)\\ \sin(\theta/2)\end{bmatrix} \qquad \cos\left(\tfrac{\theta}{2}\right)|0\rangle - \sin\left(\tfrac{\theta}{2}\right)|1\rangle = \begin{bmatrix}\cos(\theta/2)\\ -\sin(\theta/2)\end{bmatrix}, \tag{2.71}$$

which happen to also be vectors in $\mathbb{R}^2$, and which lie at an angle of $\pm\theta/2$ from the vector $|0\rangle$. This is still true if you allow yourself to consider values $0 \leqslant \theta \leqslant 2\pi$ as well: while this may no longer be a polar angle on the Bloch sphere, $\theta/2$ would still represent a meaningful angle between vectors in $\mathbb{C}^2$. These state-vectors describe a 'great circle' on the Bloch sphere, where it meets the plane spanned by the $x$ axis and the $z$-axis. In particular, the two states at angles of $\theta = \pi/2$ on the Bloch sphere are the states that lie on the $x$-axis (for which we define special symbols '$|+\rangle$' and '$|-\rangle$'):

$$|+\rangle := \tfrac{1}{\sqrt{2}}|0\rangle + \tfrac{1}{\sqrt{2}}|1\rangle = \begin{bmatrix}1/\sqrt{2}\\ 1/\sqrt{2}\end{bmatrix} \qquad |-\rangle := \tfrac{1}{\sqrt{2}}|0\rangle - \tfrac{1}{\sqrt{2}}|1\rangle = \begin{bmatrix}1/\sqrt{2}\\ -1/\sqrt{2}\end{bmatrix}. \tag{2.72}$$

These two states have $|0\rangle$ and $|1\rangle$ coefficients which have the same magnitude of $\frac{1}{\sqrt{2}}$: if we perform a standard basis measurement on either of these states, we obtain the outcomes 0 and 1 each with probability $|\frac{1}{\sqrt{2}}|^2 = \frac{1}{2} = 0.5$. Yet they are perfectly distinguishable: they represent points on opposite sides of the $x$-axis and we have $\langle +|-\rangle = \frac{1}{2}\langle 0|0\rangle - \frac{1}{2}\langle 1|1\rangle = 0$ (the way that we would distinguish them is, naturally, with an $x$-axis measurement: we will describe this further below).

**States on the equator of the Bloch sphere**
We see that there is more than one state which can be described as 'half-way' between $|0\rangle$ and $|1\rangle$. In fact, this is true of every state on the equator of the Bloch sphere – states with angular coordinates $(\pi/2, \phi)$ for all possible azimuthal angles $-\pi < \phi \leqslant \pi$. (The states $|+\rangle$ and $|-\rangle$ have $\phi = 0$ and $\phi = \pi$ respectively.) Any one of these states should produce the outcomes $r \in \{0, 1\}$ with equal probability if we perform a standard basis measurement on them, which means that they must all be described by state-vectors

$$|\psi\rangle = \begin{bmatrix}a\\ b\end{bmatrix}, \qquad \text{where } |a| = |b| = 1/\sqrt{2}. \tag{2.73}$$

For such states, both $a$ and $b$ must take the form $\frac{1}{\sqrt{2}}\mathrm{e}^{i\alpha} \in \mathbb{C}$ for some angle $\alpha \in \mathbb{R}$. In fact, we may describe the states on the equator of the Bloch sphere by vectors

$$|\psi\rangle \;=\; \tfrac{1}{\sqrt{2}}\,|0\rangle \;+\; \tfrac{1}{\sqrt{2}}\,\mathrm{e}^{i\phi}\,|1\rangle \;=\; \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ \mathrm{e}^{i\phi}\end{bmatrix}, \tag{2.74}$$

where $\phi$ is the azimuthal angle of the state on the Bloch sphere. For example, we can describe the two states on the $y$-axis by taking $\phi = \pm\frac{\pi}{2}$, for which we define the special notation '$|{+i}\rangle$' and '$|{-i}\rangle$':

$$|{+i}\rangle \;:=\; \tfrac{1}{\sqrt{2}}\,|0\rangle \;+\; \tfrac{i}{\sqrt{2}}\,|1\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\ i\end{bmatrix} \qquad\qquad |{-i}\rangle \;:=\; \tfrac{1}{\sqrt{2}}\,|0\rangle \;-\; \tfrac{i}{\sqrt{2}}\,|1\rangle = \tfrac{1}{\sqrt{2}}\begin{bmatrix}1\\ -i\end{bmatrix}. \tag{2.75}$$

**Single-qubit states in general**
If we consider the states that lie on the equator of the Bloch sphere, and then vary the polar angle $\theta$, we obtain the range of all possible single-qubit states. Combining the observations we've made above, we have the following:

**Proposition:** A single-qubit state, described by the angular coordinates $(\theta,\phi)$ on the Bloch sphere (with polar angle $\theta$ and azimuthal angle $\phi$), is represented by the state-vector

$$|\psi\rangle \;=\; \cos\!\big(\tfrac{\theta}{2}\big)\,|0\rangle \;+\; \mathrm{e}^{i\phi}\sin\!\big(\tfrac{\theta}{2}\big)\,|1\rangle \;=\; \begin{bmatrix}\cos(\theta/2)\\ \mathrm{e}^{i\phi}\sin(\theta/2)\end{bmatrix}. \tag{2.76}$$

In the above, the $|0\rangle$ component of $|\psi\rangle$ is always real-valued (and in fact non-negative). We will often prefer to represent single-qubit states this way, but it is perfectly valid to consider states where the $|0\rangle$ component is negative, imaginary, or complex-valued in general. To describe the meaning of this, we first wish to discuss measurements.

### 2.4.4 Single-qubit measurements and distinguishability

For any single-qubit state $|\beta\rangle = u\,|0\rangle + v\,|1\rangle \in \mathbb{C}^2$, the state $|\beta'\rangle = v^*\,|0\rangle - u^*\,|1\rangle$ is orthogonal to it, as $\langle\beta'\,|\beta\rangle = 0$. Then $|\beta\rangle\,, |\beta'\rangle$ form an **orthonormal basis** for $\mathbb{C}^2$: this just means that the two states are orthogonal unit vectors, which we can use to decompose any other vector $|\psi\rangle \in \mathbb{C}^2$ in some combination $|\psi\rangle = c\,|\beta_0\rangle + d\,|\beta_1\rangle$, where in particular we will have $|c|^2 + |d|^2 = 1$. We can use this to describe a sort of measurement, which perfectly distinguishes the two states $|\beta\rangle$ and $|\beta'\rangle$.

**Definition 10** (Single-qubit measurements)**.** Let $|\beta_0\rangle\,, |\beta_1\rangle \in \mathbb{C}^2$ be an orthonormal basis for $\mathbb{C}^2$. Then a $|\beta_0\rangle\,, |\beta_1\rangle$-basis measurement is a process which takes a qubit in some state $|\psi\rangle \in \mathbb{C}^2$ as input, and produces an outcome $r \in \{0,1\}$ as output, according to the probability distribution

$$\Pr\!\big[r\!=\!0\big] = |\langle\beta_0\,|\psi\rangle\,|^2, \qquad\qquad \Pr\!\big[r\!=\!1\big] = |\langle\beta_1\,|\psi\rangle\,|^2. \tag{2.77}$$

Afterwards, the qubit itself is in the state $|\beta_0\rangle$ if $r\!=\!0$, and the state $|\beta_1\rangle$ if $r\!=\!1$.

A standard basis measurement is just the special case of $|\beta_0\rangle = |0\rangle$, and $|\beta_1\rangle = |1\rangle$. We define three special sorts of measurements, involving the vectors defined in Eqns. (2.72) and (2.75):

**Definition 11** ($x$-, $y$-, and $z$-axis measurements)**.** We define:

- an $x$-axis measurement to be a measurement with respect to the $|+\rangle, |-\rangle$ basis (corresponding to states on the $x$-axis of the Bloch sphere);
- a $y$-axis measurement to be a measurement with respect to the $|+i\rangle, |-i\rangle$ basis (corresponding to states on the $y$-axis of the Bloch sphere); and
- a $z$-axis measurement to be a measurement with respect to the $|0\rangle, |1\rangle$ basis (corresponding to states on the $z$-axis of the Bloch sphere).

There is an interesting consequence to the definition of measurement described above; for an orthonormal basis $|\beta\rangle = u\,|0\rangle + v\,|1\rangle$ and $|\beta'\rangle = v^*\,|0\rangle - u^*\,|1\rangle$, the state-vector $|\psi\rangle = \mathrm{e}^{i\alpha}\,|\beta\rangle$ can't be distinguished in any way from the state-vector $|\beta\rangle$.

- For a measurement in the $|\beta\rangle, |\beta'\rangle$ basis, the result would be exactly the same as for the state-vector $|\beta\rangle$, as
$$|\langle\beta\,|\psi\rangle|^2 \;=\; |\mathrm{e}^{i\alpha}\,\langle\beta\,|\beta\rangle|^2 \;=\; |\mathrm{e}^{i\alpha}|^2 \;=\; 1. \tag{2.78}$$
If we associate $|\beta\rangle$ with the outcome '0', then the result of performing this measurement on $|\psi\rangle$ would be '0', with certainty.
- For any other measurement, e.g., with respect to some orthonormal basis $|\delta\rangle, |\delta'\rangle$, we have the same measurement probabilities when performing a measurement on the state-vector $|\psi\rangle$, as on a qubit in the state given by $|\beta\rangle$, because
$$|\langle\delta\,|\psi\rangle|^2 \;=\; |\,\mathrm{e}^{i\alpha}\,\langle\delta\,|\beta\rangle\,|^2 \;=\; |\mathrm{e}^{i\alpha}|^2 \cdot |\langle\delta\,|\beta\rangle|^2 \;=\; |\langle\delta\,|\beta\rangle|^2\,. \tag{2.79}$$

In fact: the two state-vectors $|\beta\rangle$ and $|\psi\rangle = \mathrm{e}^{i\alpha}\,|\beta\rangle$ represent the same state: the factor of $\mathrm{e}^{i\alpha}$ is an artefact of the way we represent these states by vectors in $\mathbb{C}^2$, and does not correspond to any measurable physical property. We saw this previously, when we mentioned that $|0\rangle$ and $-\,|0\rangle$ represent the same state (corresponding to the north pole of the Bloch sphere); similarly, we can say that $-\frac{1}{\sqrt{2}}\,|0\rangle + \frac{1}{\sqrt{2}}\,|1\rangle$ represents the same state as $|-\rangle$, as for that matter does $\frac{i}{\sqrt{2}}\,|0\rangle - \frac{i}{\sqrt{2}}\,|1\rangle$.

We refer to such a scalar factor difference between state-vectors, as a **global phase**. This motivates the following observation, which relates any single-qubit state-vector to the representation described in Eqn. (2.76) motivated by the Bloch sphere:

**Proposition:** Let $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$. There is an angle $0 \leqslant \theta \leqslant \pi$ such that $\cos\!\left(\frac{\theta}{2}\right) = |a|$, an angle $-\pi < \alpha \leqslant \pi$ such that $a = \mathrm{e}^{i\alpha}\cos\!\left(\frac{\theta}{2}\right)$, and another angle $-\pi < \phi \leqslant \pi$ such that $b = \mathrm{e}^{i(\alpha+\phi)}\sin\!\left(\frac{\theta}{2}\right)$. We may then write
$$|\psi\rangle \;=\; a\,|0\rangle + b\,|1\rangle \;=\; \mathrm{e}^{i\alpha}\Big(\cos\!\left(\tfrac{\theta}{2}\right)|0\rangle + \mathrm{e}^{i\phi}\sin\!\left(\tfrac{\theta}{2}\right)|1\rangle\Big). \tag{2.80}$$

We refer to $\mathrm{e}^{i\alpha}$ as a **global phase**, and to $\mathrm{e}^{i\phi}$ as a **relative phase**.

## Summary

Well done! You have now reached the end of this week's Study – Week 2: Algebra and geometry, normal matrices and introducing quantum information.

We have made a first exploration of quantum computing in the form of single-qubit states and measurements. Next week, we will build on this to consider how we can describe simple quantum computations.

This content has been prepared by Niel De Beaudrap and Hex Miller-Bakewell on behalf of the University of Sussex.